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A hybrid data envelopment analysis and artificial intelligence framework for sustainable supplier selection: a case study in the petrochemical industry

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ABSTRACT

As global industrial operations expand, the complexity and volume of suppliers have increased, making sustainable supplier selection (SSS) a strategic imperative for resilient supply chains (SCs). Traditional evaluation methods often fail to handle large datasets and dynamic sustainability metrics, resulting in suboptimal decisions. This study introduces a novel hybrid intelligent framework that integrates Most Productive Scale Size Data Envelopment Analysis (MPSS-DEA) with three Artificial intelligence (AI) algorithms—Artificial Neural Networks (ANNs), K-Nearest Neighbors (KNN), and Chi-squared Automatic Interaction Detection (CHAID)—to enhance both accuracy and scalability in supplier evaluation. Applied to 362 suppliers and 38 sustainability criteria in the petrochemical industry, the proposed framework achieved high classification accuracies of 92.5% (DEA-CHAID), 91.9% (DEANN), and 91.4% (DEA-KNN). The model also demonstrated strong discrimination power, with ROC AUC scores of 0.96, 0.95, and 0.94, respectively. Predictor importance analysis revealed that some of the features, such as R2, SE2, QA4, CA6, and DE3, were the most influential features across all models. Beyond performance metrics, the framework offers real-time supplier replacement, reduced computational complexity, and modular adaptability across industries. It supports ethical and sustainable sourcing by integrating economic, environmental, and social dimensions into decision-making. The intelligent architecture enables lifecycle analysis, promotes transparency, and aligns with global sustainability standards. This research contributes a scalable, interpretable, and data-driven solution for sustainable supplier selection, bridging the gap between traditional DEA models and modern artificial intelligence (AI) techniques. Its applicability across diverse industrial contexts positions it as a robust tool for strategic procurement and supply chain resilience.

1. Introduction

Fig. 1 presents the graphical abstract of the proposed hybrid intelligent framework for SSS in the petrochemical industry. It visualizes the integration of MPSS-DEA with three AI algorithms—ANN, KNN, and CHAID—alongside comparative performance metrics. Among the models, DEA-CHAID achieved the highest accuracy (92.5 %), underscoring the framework's robustness in managing large-scale datasets and complex sustainability criteria.

To enhance clarity and accessibility for readers unfamiliar with technical abbreviations and domain-specific terms, Table 1 provides a comprehensive nomenclature. It summarizes all key symbols, acronyms, and variables used throughout the manuscript, particularly those related to DEA modeling, ML algorithms, and supplier evaluation metrics.

As environmental concerns and corporate social responsibility continue to rise globally, organizations are under increasing pressure to adopt sustainable practices throughout their SCs[23]. Procurement plays a pivotal role in maintaining SC competitiveness by ensuring timely delivery and high-quality products. A critical component of procurement is supplier selection (SS), which directly influences operational success and long-term strategic outcomes. Poor supplier choices can lead to delays, financial losses, and reputational damage[32].

SSS has emerged as a key process for evaluating suppliers based on environmental, social, and economic criteria. However, traditional evaluation methods often rely on subjective judgments and limited data, rendering them inadequate for today's data-intensive decision-making environments [37]. As industries face increasingly complex SCs and large volumes of supplier data, the need for advanced analytical tools

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becomes more urgent[22]. Recent developments in AI and machine learning (ML) have shown great promise in addressing these challenges. AI-based models can enhance SSS by incorporating predictive analytics, identifying patterns, and integrating sustainability metrics into decision-making processes[34]. These technologies enable organizations to assess suppliers more holistically, optimize resource allocation, and improve transparency and resilience across the SC[40].

Generally, the process of SSS can be structured into several key stages[3]:

1. Understanding the strategic goals and operational priorities of the target industry.
2. Defining appropriate criteria for evaluating suppliers within the industry context.
3. Assigning relative importance and weights to each criterion
4. Assessing the capabilities and strengths of potential suppliers.
5. Ranking suppliers based on their performance against the defined criteria.
6. Selecting one or more top-ranked suppliers as optimal choices.

Numerous studies across various industries have employed traditional multi-criteria decision-making (MCDM) techniques, AI, ML, or hybrid combinations of these approaches to address SSS. Each method offers distinct advantages and limitations, which can be effectively mitigated through strategic integration[1,6,46,60,64,65].

Despite these advancements, both MCDM methods and ML techniques face limitations. MCDM approaches often struggle with high-dimensional criteria, data sensitivity, and computational complexity. ML models, on the other hand, are prone to overfitting, data bias, and lack of interpretability—especially when applied to high-dimensional, multi-class classification problems. Most prior studies rely on small datasets, limited criteria (typically fewer than 20), and static evaluation frameworks that fail to adapt to the dynamic nature of sustainability metrics. Moreover, traditional DEA models, while effective in multi-criteria efficiency analysis, require full recalculation when new suppliers are introduced, making them impractical for large-scale, real-time decision-making. From the standpoint of international research and practical implementation, there is a clear need for a scalable, intelligent framework that can handle high-dimensional data, reduce computational burden, and support dynamic supplier evaluation.

The problem statement guiding this research is as follows: How can sustainable suppliers be efficiently selected and managed in large-scale,

Table 1

List of abbreviations and symbols used in the study.

Symbol / Term	Description
SSS	Sustainable Supplier Selection
SS	Supplier Selection
MPSS-DEA	Most Productive Scale Size Data Envelopment Analysis
ANN	Artificial Neural Network
KNN	K-Nearest Neighbors
CHAID	Chi-squared Automatic Interaction Detection
ML	Machine Learning
AI	Artificial Intelligence
DMU	Decision Making Unit (e.g., supplier)
SCM	Supply Chain Management
e_o	Efficiency score output from DEA
MCDM	Multi-Criteria Decision Making
CFA	Confirmatory Factor Analysis
SEM	Structural Equation Modeling
FDEMATEL	Fuzzy Decision-Making Trial and Evaluation Laboratory
ANP	Analytic Network Process
DT	Decision Tree
NB	Naive Bayes
ROC	Receiver Operating Characteristic
AUC	Area Under Curve (ROC metric)
TOPSIS	Technique for Order of Preference by Similarity to Ideal Solution
AHP	Analytic Hierarchy process
GRA	Grey Relational Analysis
BWM	Best-Worst method
MLP	Multi-Layer Perceptron

data-rich industrial environments, while minimizing computational burden and maximizing decision accuracy across economic, environmental, and social criteria?

To answer this question and address these gaps, this study introduces a novel hybrid intelligent framework that integrates the MPSS-DEA with three AI algorithms—ANNs, KNN, and CHAID. The innovation lies in the synergy between DEA's robust efficiency evaluation and the predictive power of AI-ML methods, enabling accurate classification of suppliers into efficient and inefficient categories, while significantly reducing computational time. Unlike previous studies, our framework is designed to handle large datasets with high-dimensional sustainability criteria. Specifically, we apply the model to a real-world case involving 362 suppliers and 38 criteria in the petrochemical industry—one of the most data-intensive and strategically critical sectors in global SCs.

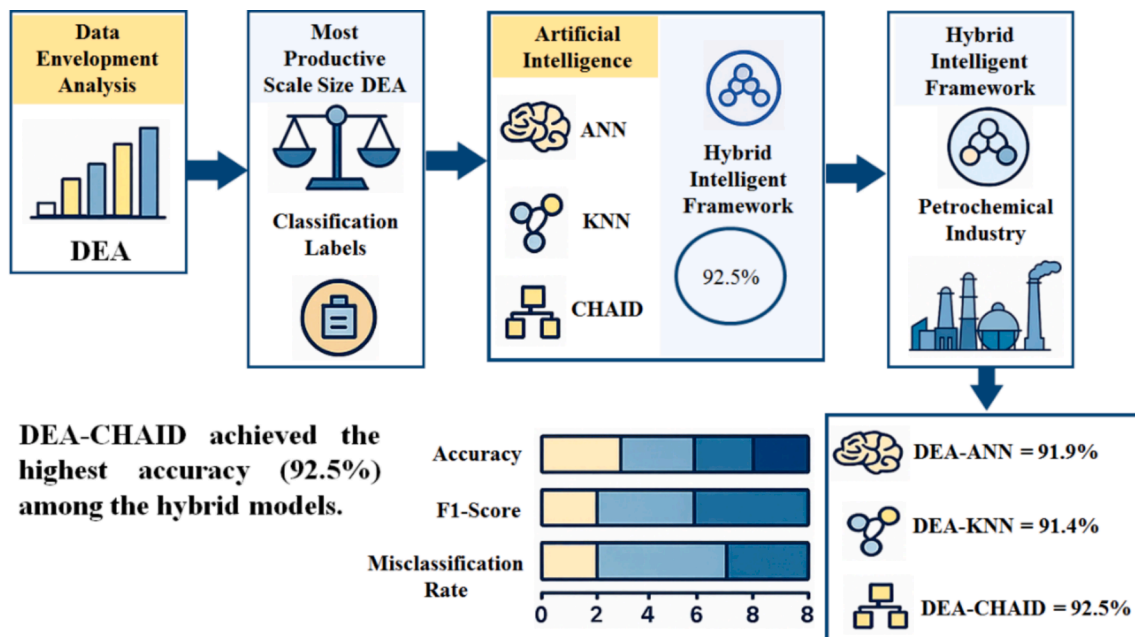


Fig. 1. Graphical abstract of the hybrid DEA-AI framework and comparative performance of models.

The results demonstrate the superiority of the proposed hybrid models in terms of classification accuracy and robustness. DEA-CHAID achieved the highest accuracy at 92.5 %, followed by DEA-ANN at 91.9 % and DEA-KNN at 91.4 %, based on confusion matrix metrics including precision, recall, and $F1$ – score. Additionally, predictor importance analysis revealed that environmental and compliance-related criteria had the greatest influence on supplier classification, highlighting the model's ability to uncover meaningful patterns in complex datasets. The integration of MPSS-DEA with AI not only improves decision-making quality but also enables real-time supplier replacement, risk assessment, and strategic planning.

From the standpoint of international research and industrial practice, this study advances the state-of-the-art by offering a scalable, adaptable, and high-performance solution for SSS. It contributes both methodologically and practically by bridging the gap between efficiency analysis and intelligent classification, and by demonstrating its applicability in a large-scale, real-world industrial setting.

In summary, this study follows these objectives including:

1. Develop a hybrid intelligent framework for SSS by integrating MPSS-DEA with AI algorithms (ANN, KNN, and CHAID).
2. Evaluate supplier efficiency and classify them into MPSS and non-MPSS categories using DEA.
3. Train and validate AI models using labeled data and assess their performance through confusion matrices and ROC curves.
4. Identify the most influential criteria for supplier sustainability across economic, environmental, and social dimensions.
5. Demonstrate the applicability of the proposed method in large-scale industrial settings, particularly within the petrochemical sector.

Moreover, this study makes the following key contributions:

1. Development of a hybrid intelligent framework combining MPSS-DEA with ANN, KNN, and CHAID to enhance supplier evaluation accuracy and scalability.
2. Reduction of computational complexity in DEA, enabling faster and more dynamic supplier assessments.
3. Handling high-dimensional and large-scale datasets, applied to 362 suppliers and 38 criteria.
4. Identification of key sustainability criteria through predictor importance analysis.
5. Real-world validation in the petrochemical industry, demonstrating practical applicability and high accuracy.
6. Contribution to international research and industrial practice, offering a scalable, adaptable, and data-driven solution for sustainable supplier management.

Fig. 2 illustrates the conceptual flowchart of this study.

The remainder of the paper is structured as follows: [Section 2](#) reviews related works concerning SS and petrochemical keywords. [Section 3](#) presents the proposed methodology. [Section 4](#) outlines the case study, detailing each method and the corresponding results. [Sections 5 and 6](#) provide the discussion and conclusion, respectively.

2. Related works

The methods mentioned in the introduction are divided into three categories: (i) MCDM methods for SS, (ii) AI algorithms for SS, and (iii) integrated methods based on MCDM and AI for SS. Within these categories, the strengths and weaknesses of each method are identified. The

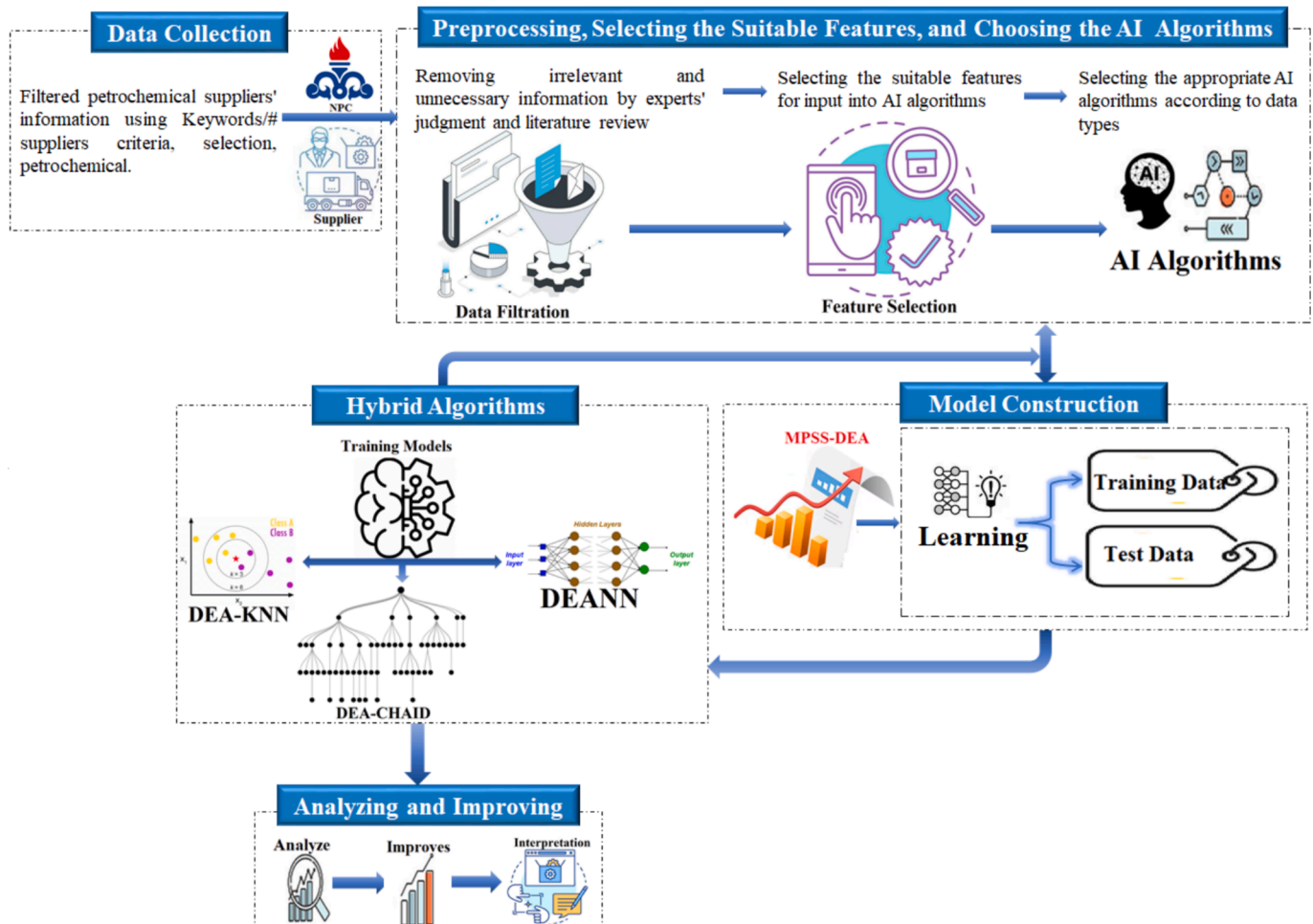


Fig. 2. The conceptual flowchart of this study.

primary reason for selecting the integrated approach in this study is to minimize the limitations associated with individual methods. Before delving into the literature review, Table 2 presents some of the key criteria frequently used in various studies.

2.1. MCDM methods for supplier selection

Identifying and evaluating appropriate suppliers is crucial to the success of any organization. The SS process involves several key objectives, such as minimizing purchasing risk, ensuring customer satisfaction, and fostering long-term relationships between buyers and suppliers. Various methods-including AHP, Analytic Network Process (ANP), DEA, TOPSIS, and ELECTRE- are utilized to assess suppliers and identify the best options[56]. Bafrooei et al. [11] proposed a combined DEA-MCDM model to rank suppliers in Iran's petrochemical industry. They introduced a new criterion, "executive capability," based on their own experiments and expert opinions. The findings revealed the significant importance of this criterion in SS within the petrochemical sector. Sayfi and Nikbakht [51] proposed an integrated approach to identify and prioritize criteria for selecting green suppliers in the petrochemical industry. They used a T-test to identify relevant criteria and employed the FANP method for supplier ranking.

Prasad and Kambagowni [46] proposed a combined approach using DEA, AHP, and GRA to evaluate and select the best suppliers for companies. They applied this method in a steel plant in southern India, demonstrating its effectiveness for both qualitative and quantitative criteria. He and Zhang [26] introduced an integrated framework based

on factor analysis (FA), DEA, and AHP to select suitable suppliers for cement firms. The results indicated that the proposed method could comprehensively evaluate and rank efficient suppliers. Zarbakhshnia and Jamali Jaghdani [64] and Eydi and GholamAzad [20] developed a novel NDEA and NEREC- NDEA models that considers uncontrollable and undesirable outputs for SS in Iran's plastic industry. Their findings showed that the model possesses reliable and valid characteristics, making it suitable for application in similar industries. Tong et al. [57] presented a valuable approach using FTOPSIS for assessing equipment maintenance vendors in the chemical sector. They employed eight primary criteria and twenty-four sub-criteria to select appropriate suppliers, ensuring future operational safety and sustainability. The findings demonstrated the method's significant practical utility for adoption in chemical industries. Wongthong and Paoprasert [60] introduced a novel SS approach for maintenance tasks in Thailand's petrochemical sector, utilizing confirmatory factor analysis (CFA) and structural equation modeling (SEM). Their findings indicated that factors such as cost, quality, flexibility, and human resources greatly influence SS. Tong et al. [58] proposed a new framework for selecting and evaluating suitable suppliers for a Chinese petrochemical company using the fuzzy PROMETHEE II method. This approach was structured around three dimensions: fundamental, procedural, and sustainability factors. The results highlighted the method's efficacy in SS.

Karami et al. [33] proposed a three-stage hybrid method- PCA-DEA-VIKOR- to address SS issues in garment SCs. The study demonstrated the effectiveness of this integrated approach for application in similar SCs. Mina et al. [38] introduced a new approach using FIS-FAHP-FTOPSIS for

Table 2
The numerous criteria for supplier selection.

	[11]	[51]	[46]	[26]	[64]	[57]	[60]	[58]	[33]	[38]	[42]	[6]	[1]	[3]	Mai et al., 2019	Soheilzadeh & Zargaran khouzani, 2022
Cost	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓	✓		
Quality	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓		
On time delivery	✓✓		✓	✓			✓		✓	✓		✓	✓	✓		
Executive capabilities	✓			✓					✓	✓						
Risk	✓					✓								✓		
Supplier qualification				✓				✓								
Maintenance service performance				✓		✓		✓								
Schedule control								✓								
Site management								✓								
Sub-supplier management							✓	✓		✓						✓
Environmental sustainability		✓		✓	✓			✓	✓	✓	✓		✓	✓		
Social sustainability								✓	✓							
Warranties and claim policies						✓										
Price		✓	✓	✓		✓			✓		✓	✓		✓	✓	
Mutual trust and easy communication			✓						✓							
Communication system				✓					✓						✓	✓
Labor relations record				✓	✓	✓	✓		✓							
Geographical location		✓											✓		✓	
Process improvement				✓												
Innovativeness											✓	✓				
Green products		✓														
Flexibility		✓					✓			✓	✓		✓			✓
Reliability		✓	✓	✓	✓		✓		✓	✓					✓	
Transportation cost												✓				✓
Quantity discount		✓								✓		✓		✓		
Service		✓				✓✓			✓			✓	✓		✓	
Technology capability		✓									✓		✓			
Occupational health and safety system					✓				✓		✓	✓	✓			
Performance history				✓									✓			✓
Customer Satisfaction		✓								✓	✓					
Capability			✓	✓					✓	✓	✓			✓	✓	
Corporate social responsibilities (CSR)															✓	

SS and evaluation in the circular SC. Their study focused on six petrochemical companies in Iran, showing that the proposed framework offers greater efficiency and applicability compared to existing methods. Nguyen et al. [42] suggested a new integrated method based on DEA-SF-AHP and SF-WASPAS to identify and select sustainable suppliers in Vietnam's steel manufacturing sector. The findings indicated that the proposed integrated method is a practical and robust approach to SS and can be effectively utilized in the context of big data for the steel industry.

Gidiagba et al. [23] developed an integrated MCDM method for selecting suppliers in the oil and gas industry. They utilized the Best-Worst method (BWM) to determine the main criteria and ranked the suppliers using TOPSIS. Fifteen criteria were selected in this study, incorporating the Delphi technique. Nasri et al. [43] introduced an integrated FDEMATEL-ANP-DEA method to identify sustainable suppliers in Iranian petroleum companies. This method enables managers to reduce adverse environmental impacts and enhance the social aspects of the SC. Parsa Rad et al. [47] presented an integrated MCDM method using the fuzzy Best-Worst Method (FBWM) and the Gray Combined Compromise Solution (COCOSO) for SS in oil, gas, and petrochemical companies. The study aims to improve SS under uncertainty. The findings indicate that the method expedites and streamlines the selection process in this industry.

2.2. AI methods for supplier selection

In recent years, numerous researchers have increasingly utilized ML and AI techniques to select sustainable suppliers. In this context, Omurca [45] introduced a novel approach using a fusion of fuzzy c-means (FCM) and rough set theory (RST) for SS and evaluation. The findings demonstrated that this method can identify top suppliers and categorize all vendors according to their fuzzy similarity levels. Guo et al. [21] proposed a new semi-fuzzy SVDD model to address the SS multi-classification issue. The study segmented observations into deterministic and fuzzy samples, enabling the method's application under uncertain conditions. This approach not only streamlines the calculation process but also enhances supplier performance. El-Hiri et al. [19] proposed an ANN-based method to reduce supply and demand risks, assisting manufacturers in selecting the most efficient suppliers. This approach enables managers to make appropriate supplier choices without relying on complex algorithms. Jafari Nodeh et al. [30] introduced an intelligent supervised approach employing Multi-Layer Perceptron (MLP), Decision Tree (DT), and K-means to select and evaluate suppliers in the oil, gas, and petrochemical sectors. This method reduces runtime and costs while effectively ranking top suppliers. Zhao et al. [65] presented a novel SVM-based approach to address SS challenges. Their findings indicated that the method is effective in categorizing expert data, assessing historical information, and assigning weights. Ali et al. [6] presented a random forest approach based on supervised ML for SS. This method can generate a detailed list of key criteria commonly used in supplier evaluation, allowing managers to focus on criteria most relevant to their suppliers. Abulouifa and Bahj [7] proposed an integrated framework using ML algorithms such as AdaBoost, SVM, and KNN to select suppliers based on their orders and preferences. The findings revealed that the AdaBoost algorithm achieved the highest accuracy (98 %) compared to the others, with an *F1*-score of 86 %. Abdelhadi and Almomani [8] examined similarities among supplier groups using specific criteria. They introduced a novel clustering technique, Crisp Gradual Means Integral, to assess supplier similarity and assign each to a distinct category. The results indicated that this method can help managers avoid adverse situations by replacing unavailable suppliers with those possessing comparable attributes. Islam et al. [29] introduced a three-phase framework for addressing SS and order allocation. By utilizing Seasonal Autoregressive Integrated Moving Average (SARIMA) and Light Gradient-Boosting Machine (LGBM), they predicted customer demand and compared the outcomes of both methods. Additionally, a deep network was established to recommend optimal choices

for suppliers in the Canadian meat industry. The findings revealed significant correlations among variables in SS. Mirzaee and Ashtab [41] conducted a comprehensive review of SS, focusing on sustainability, resiliency, and AI-based methods. Their study analyzed 143 research articles and covered various evaluation criteria, including economic, environmental, social, and resilience aspects. They explored multiple hybrid approaches, such as mathematical models, MCDM-ML techniques, and AI algorithms. The study culminated in an integrated framework and identified seven potential directions for future research in this evolving field. Asghar et al. [9] employed an intelligent computing framework using the Levenberg–Marquardt algorithm (LMA) to train artificial neural networks for accurate prediction of fluid forces in transient non-Newtonian thermal flows. By integrating CFD-generated data with ANN-LMA, the model achieves high precision while significantly reducing computational cost. The approach offers a robust alternative to traditional simulations in complex engineering fluid dynamics. Yadav et al. [62] proposed an optimization method combining Mixed-Integer Nonlinear Programming (MINLP) and ML to optimize digital procurement costs for SS, while ensuring transparency, traceability, security, and comprehensive information on a distributed ledger. This method is primarily applied to manage SCs for medical equipment, drugs, liquid oxygen, and other essential goods during epidemics. Shafiq Hira et al. [53] presented a computational framework combining CFD simulations with artificial neural networks (ANN) to analyze Darcy–Forchheimer flows in stenosed arteries. By modeling nonlinear fluid behavior in porous, constricted vessels, the ANN-CFD coupling enables accurate prediction of hemodynamic forces. The approach supports medical diagnostics and treatment planning by reducing simulation time while maintaining precision and accuracy. Gidiagba et al. [24] conducted a systematic review on the application of ML for SSS between 2010 and 2024. By analyzing 50 key studies from major databases, the authors demonstrated how ML techniques improve decision-making by managing complex, multi-dimensional data and addressing the shortcomings of traditional approaches. Their proposed framework aligns AI and conventional methods across all stages of supplier evaluation-criteria definition, weighting, assessment, and ranking- resulting in enhanced accuracy, efficiency, and organizational performance.

2.3. Methods of integrating MCDM and AI

Since each method for SS may have its own weaknesses, many researchers have utilized a combination of various approaches to assess and select suppliers. For instance, Tavana et al. [54] introduced a novel ANN-based method and ANFIS to understand the relationships among key features in SS. In their study, the AHP method was employed to determine the most influential criteria for evaluating supplier performance. The proposed method was tested in the automotive sector to demonstrate its robustness and responsiveness. Cheng et al. [17] presented an integrated DEA-Adaboost model for SS. Fuzzy multi-objective DEA was used to construct the expert dataset, and the Adaboost approach was applied to train the dataset for the intelligent model. The findings showed that the proposed model achieved higher accuracy compared to other models. Amindoust [1] proposed a resilient-sustainable method based on fuzzy set theory, fuzzy inference system (FIS), and AR-DEA to intelligently select suppliers. This integrated approach presents a comprehensive SS model with multiple criteria. The method was implemented in a case study of Imen Soukht Sepahan, and sensitivity analysis was conducted. Ijadi Maghsoodi et al. [28] introduced a new method for SS using the MULTIMOORA technique and k-means cluster analysis, demonstrating a fast and efficient process based on distinct clusters. Abdulla et al. [2] proposed a hybrid MCDM-ML method utilizing AHP and DT for effective SS and evaluation. The method was applied to Libyan companies to highlight the DT's lower complexity compared to AHP. Cheng et al. [18] presented an intelligent model for selecting and evaluating suppliers in global SCs using data-

driven Support Vector Regression (SVR). In this context, the MCDM method was applied to determine supplier labels, and SVR was utilized for model training. The findings indicated that the proposed model demonstrated strong reliability and accuracy. Aggarwal et al. [5] proposed a novel model combining DEA with SVM, DT, and logistic regression for supplier selection. The results identified DT as the most accurate method. Zhu et al. [66] introduced a novel approach integrating the DEA method with ML algorithms to evaluate supplier efficiency. They employed Back Propagation Neural Networks (BPNs), ISVM, and Generative Adversarial Neural Network (GANN) algorithms alongside DEA in Chinese manufacturing firms. The findings showed a 94 % average efficiency improvement using hybrid techniques, surpassing the performance of the standalone DEA model. Abdulla et al. [3] proposed a combined method based on the Measurement of Alternatives and Ranking according to Compromise Solution (MARCOS) and tree-based ML for evaluating and selecting sustainable suppliers. In their study, the importance of each criterion was determined using a decision tree and applied within the MARCOS method to rank suppliers in the oil and gas industry. They claimed that the proposed integrated method could be useful for designing an efficient and effective SS process. Liang et al. [35] introduced a hybrid framework combining MCDM and AI techniques to facilitate the effective integration of AI technology in SCM. They applied DEMATEL to identify relationships among criteria and used a clustering algorithm to classify automobile SC suppliers. The findings indicated the method's applicability to other industries facing similar challenges. Ma and Li [40] proposed a conceptual framework consisting of three modules: the decision matrix and criteria initialization module, the MCDM method selection and implementation module, and the aggregation module. A key focus of their study is the development of an intelligent core for decision support systems (DSSs) within the aggregation module, utilizing ML methods. This integrated approach not only evaluates supplier quality but also has broader applications. They tested their method in a large automotive company, demonstrating that the proposed DSS delivers more robust and reliable evaluation results compared to traditional MCDM methods. Abdulla and Baryannis [4] introduced a SS method that integrates interpretable, data-driven AI techniques with MCDM approaches. This study aims to simplify SS challenges while maintaining stakeholder familiarity by keeping MCDM central to the process. The method was validated through sensitivity analysis in oil, gas, and aerospace manufacturing companies, showing its capability and robustness compared to other methods that rely solely on ML techniques. Lin and Lu [36] developed a new analytical framework that combines inverse DEA with ML to optimize the performance of Apple's SC suppliers. Their study provided a systematic method to reduce CO₂ emissions by 30 % to 50 % using Random Forest (RF) and KNN algorithms along with IDEA for evaluating supplier performance and identifying the most sustainable choices. The proposed hybrid method outlines a module for SC stakeholders to integrate sustainability into their operations. The results serve as guidance for CO₂ reduction and help evaluate suppliers' economic performance to achieve sustainability goals. Nafei et al. [44] introduced a hybrid decision-making method combining the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) with Neuromorphic Triplets (NTs) to overcome the limitations of traditional methods in managing indeterminate information. This new approach enhances computational efficiency and accuracy. The findings indicate that the intelligent TOPSIS method improves decision accuracy and simplifies computations compared to traditional TOPSIS. Furthermore, it can be adapted to various scenarios across different research fields and industries. Gidiagba et al. [25] proposed a hybrid supplier evaluation model for the pharmaceutical sector by integrating non-negative matrix factorization, random forest, and TOPSIS. It reduces 24 criteria to 8 core dimensions, assigns data-driven weights, and ranks suppliers effectively. Results show improved accuracy, interpretability, and sustainability compared to traditional MCDM approaches. Tajally et al. [59] proposed a novel hybrid ML-based model for SS by integrating sustainability, circular

economy, and viability criteria. Using a case study in the automotive industry, it combines LGPSBWM for weighting dimensions and SWSVM for supplier classification. Key indicators such as cost, quality, and GHG emissions are identified, and the model selects Omani#01 and Iranian#07 as top suppliers, demonstrating high efficiency and reliability.

Based on the reviewed literature and numerous studies conducted on SS across various industries- including petrochemicals- a comprehensive study involving multiple criteria and sub-criteria alongside a large number of suppliers specifically within the petrochemical sector has not been identified. Therefore, this study is inspired by prior research such as Bafrooei et al. [11], who applied the MCDM-DEA method for SS in an Iranian petrochemical firm; Wongthong and Paoprasert [60], who employed CFA and SEM for supplier evaluation in Thailand's petrochemical industry; Nasri et al. [43], who used the FDEMATEL-ANP-DEA approach for petroleum companies in Iran; Jafari Nodeh (2020), who utilized MLP-DT-K-means for SS; Cheng et al. [17] and Amindoust [1], who applied hybrid models including FIS, AR, AdaBoost, and DEA; and studies by Aggarwal et al. [5], Zhu et al. [66], and Abdulla et al. [3], which integrated DEA with SVM, DT, ISVM, and BPNN.

This study proposes an intelligent hybrid method that combines the MPSS-DEA approach with MLP, CHAID, and KNN algorithms. Initially, the MPSS-DEA model is used to identify the most productive suppliers and classify observations into MPSS and non-MPSS groups. Subsequently, supplier performance is evaluated, and the most influential criteria are extracted using MLP, CHAID, and KNN.

The proposed method offers several advantages:

1. Reduced algorithm runtime when adding new observations compared to standalone MPSS-DEA;
2. Fast and timely replacement of underperforming suppliers with similar alternatives;
3. Grouping suppliers into comparable categories for quicker and more efficient selection.

3. Proposed methodology

This section presents a novel intelligent algorithm for SSS. Before discussing the main topic, it should be noted that the three terms- criteria, input and output parameters, and features- refer to the same concept. Each term is used according to its corresponding model: criteria (suppliers' questionnaires), input and output parameters (MPSS-DEA model), and features (ANNs, CHAID, and K-NN algorithms). Fig. 3 illustrates the main structure of the proposed hybrid intelligent algorithm.

The algorithm, shown in Fig. 3, comprises three phases. In the first phase, the collected datasets undergo cleansing using one of the studied preprocessing methods for dimensionality reduction. This process eliminates missing and outlier data while selecting relevant, unique, and monotonic features. All features are examined to determine their types and relationships, including both linear and nonlinear connections. The selected datasets are then imported into the MPSS-DEA model to evaluate supplier performance.

Efficiency rates are used to gauge supplier sustainability: an efficiency rate (e_o) of one indicates sustainability, while $e_o < 1$ indicates unsustainability. Based on these results, suppliers are categorized into sustainable and unsustainable groups, which are then split into training and testing datasets- typically in an 80/20 or 70/30 ratio- with the smaller portion reserved for testing and the larger for training the ML model. Upon completing this process, the second phase begins, where training datasets are imported into the selected ML models for learning. In this phase, each hybrid algorithm- DEANN, DEA-CHAID, and DEA-KNN- follows specific steps within SPSS Modeler version 18.0.

In phase three, confusion matrices, predictor importance charts, and ROC curves are extracted for each hybrid algorithm. The confusion matrix for the training dataset is generated using key metrics such as accuracy, precision, and F1-score. Additionally, the weights of features important for evaluating supplier sustainability are identified and prioritized within each hybrid algorithm. The ROC curves illustrate the

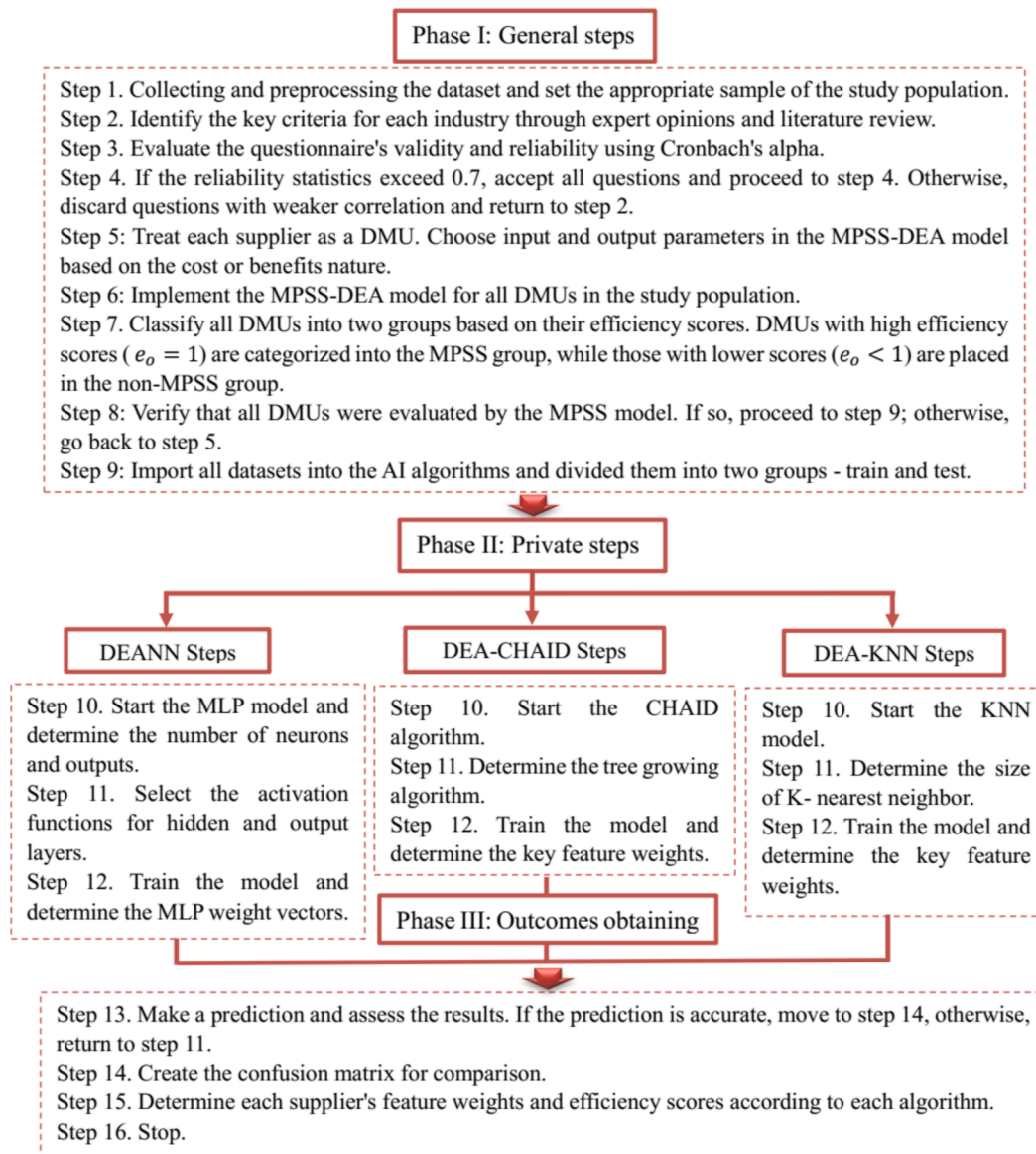


Fig. 3. The hybrid algorithm for sustainable supplier selection.

correct classification rates for each algorithm. Finally, all results are compared to identify the hybrid algorithm with the highest accuracy.

The overall structure and workflow of the proposed hybrid intelligent framework are illustrated in Fig. 4.

Based on the algorithm and models used in this study, a substantial dataset is required. The data was collected from P Petrochemical Company in Iran. Established in the late 2000 s in western Iran, P Petrochemical produces materials used in the manufacturing of feed for downstream units, polyethylene films, polyethylene foams, wire and cable coverings, and plastic components. The company currently employs approximately 800 staff across various departments and works with over 7,000 suppliers from diverse sectors, including oil and gas raw materials, polyethylene raw materials, electrical appliances, pipes, and other spare parts.

The following section outlines the key stages of the proposed algorithm and provides an explanation for each method.

4. Designing an intelligent process for sustainable supplier selection and evaluation at P petrochemical

4.1. Phase i

Step 1: Collecting and Preprocessing the Dataset and Selecting the Study Sample: The data provided by P Petrochemical is qualitative and based on expert judgment. Among the 7,000 supplier records, duplicates exist, and certain goods were supplied only once. This study employs dimension reduction for dataset preprocessing, which reduces dataset size and enables the development of a more robust AI model with shorter training times [22]. Dimension reduction is a critical step in many data analysis and ML tasks. By selecting appropriate techniques based on the dataset's characteristics and analytical goals, practitioners can enhance model performance, improve interpretability, and support effective data visualization. Linear Discriminant Analysis (LDA) was applied for data cleansing. LDA is a supervised dimension reduction technique that identifies a linear combination of features that best separates multiple classes, taking class labels into account. It is widely used in classification tasks to reduce dimensionality while preserving class-discriminatory information [50].

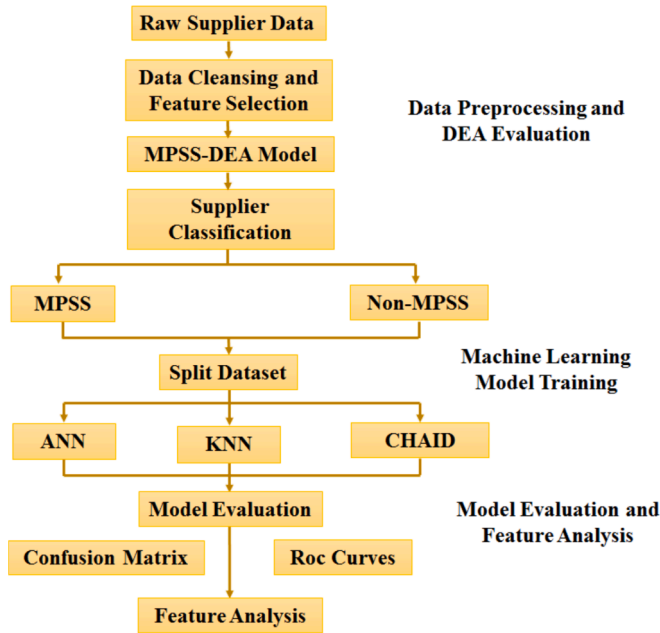


Fig. 4. Flowchart of the proposed hybrid intelligent framework for SSS.

Following these steps, 6,200 records remained. Given the complexities associated with managing a large supplier base, the Cochran equation was used to determine an appropriate sample size for the study population [12]. The Cochran formula is as follows:

$$n = \frac{\frac{z^2 pq}{d^2}}{1 + \frac{1}{N} \left[\frac{z^2 pq}{d^2} - 1 \right]} \quad (1)$$

Where n is the sample size, z is the value corresponding to the selected significance level α —for example, 1.96 for a 95% confidence level (with 0.025 in each tail). p represents the estimated proportion of an attribute present in the population, and $q = 1 - p$. The product $(p)(q)$ provides an estimate of the variance. $d = 0.05$ denotes the acceptable margin of error for the proportion being estimated. Based on these assumptions, the sample size is calculated using the following formula:

$$n = \frac{\frac{(1.96)^2 \times 0.5 \times 0.5}{(0.05)^2}}{1 + \frac{1}{6200} \left[\frac{(1.96)^2 \times 0.5 \times 0.5}{(0.05)^2} - 1 \right]} = 361.97 \approx 362$$

Based on the sample size derived from Equation (1), 400 questionnaires were distributed to assess and identify the primary criteria influencing the selection of top suppliers. Each questionnaire included a total of 38 criteria. Of the 400 distributed, 362 questionnaires with complete data were included in the final sample. The supplier selection surveys were completed by personnel from various divisions within the petrochemical sector, including procurement, general management, import and export management, and purchasing and sales teams.

Step 2. Identifying the key criteria: The selection of input parameters (criteria/features) in this study was based on a multi-step process combining expert consultation, literature synthesis (Table 2), and data relevance. Initially, a comprehensive review of prior studies in SSS—particularly in the petrochemical sector—was conducted to identify commonly used criteria across economic, environmental, and social dimensions (e.g., [11,43,66]). These criteria were then cross-validated through structured interviews with procurement and sustainability experts from the case company to ensure contextual relevance and practical applicability.

In the preprocessing phase, feature selection techniques were applied

to refine the dataset. This included removing missing and outlier values, assessing monotonicity, and ensuring uniqueness and non-redundancy of variables. Only those features that demonstrated consistent relevance across both expert judgment and statistical validation were retained. This approach ensured that the final input parameters used in the MPSS-DEA and AI models accurately reflect the multidimensional nature of supplier sustainability and are aligned with both theoretical foundations and operational realities.

Accordingly, 8 main criteria and 38 sub-criteria were selected to evaluate the most suitable suppliers for the P petrochemical company. These criteria are organized and visualized in a hierarchical flow diagram, as shown in Fig. 5.

Steps 3 and 4: Evaluating the Reliability of the Questionnaires:

Cronbach's alpha is a measure of internal consistency and scale reliability, indicating the extent to which a set of items are closely related as a group. If the reliability coefficient is 0.7 or higher, the criteria are considered to have a reliable relationship, and all questions are deemed acceptable. Otherwise, questions that reduce the overall correlation should be revised or eliminated based on expert judgment. The formula for Cronbach's alpha is as follows [13]:

$$\alpha = \frac{N\bar{c}}{\bar{v} + (N-1)\bar{c}} \quad (2)$$

N = The number of items

$\bar{c}$ = The average of inter-item covariance.

$\bar{v}$ = The average variance.

The Cronbach's alpha score for the 38 sub-criteria is reported in Table 3.

Table 3 reports an alpha coefficient of 0.80 for the 38 items, indicating that all questions exhibit reasonably strong internal consistency.

Table 4 presents the distribution of questionnaire responses for five suppliers across five criteria. The analysis indicates that the dataset follows a linear distribution.

Appendix A presents the statistical significance testing table along with the corresponding curves for one sample criterion.

Step 5: Selecting Input and Output Parameters Based on Cost or Benefit Nature: Table 5 lists the criteria selected as input and output parameters for inclusion in the MPSS-DEA model, based on a comprehensive literature review and an examination of their cost or benefit characteristics.

Step 6: Implementing the MPSS-DEA Model: DEA is a nonparametric method first introduced by Charnes et al. in 1978 to evaluate the efficiency of Decision-Making Units (DMUs) on the productivity frontier. The foundational linear mathematical model of DEA is the CCR model, named after its originators. Suppose there are n DMUs ($j = 1, \dots, n$) that consume m inputs x_{ij} , ($i = 1, \dots, m$) to produce s outputs y_{rj} , ($r = 1, \dots, s$). The CCR model based on these assumptions is formulated as follows [15]:

$$e_o = \text{Min} \left\{ \theta \left| \begin{array}{l} \sum_{j=1}^n \lambda_j x_{ij} \leq \theta x_{io}, \quad i = 1, \dots, m, \\ \sum_{j=1}^n \lambda_j y_{rj} \geq y_{ro}, \quad r = 1, \dots, s, \quad \lambda_j \geq 0 \end{array} \right. \right\} \quad (3).$$

θ = The overall efficiency

models have been developed based on the CCR framework, one of which is the MPSS model, used to calculate the productivity rate of DMUs. The MPSS model was first introduced by Banker and Kemerer in 1989. According to DEA principles, a unit is classified as MPSS when the average productivity across all input-output combinations is maximized. In simpler terms, a DMU is identified as MPSS if both its technical and scale efficiency equal 1. The mathematical formulation of the MPSS model is presented as follows [10]:

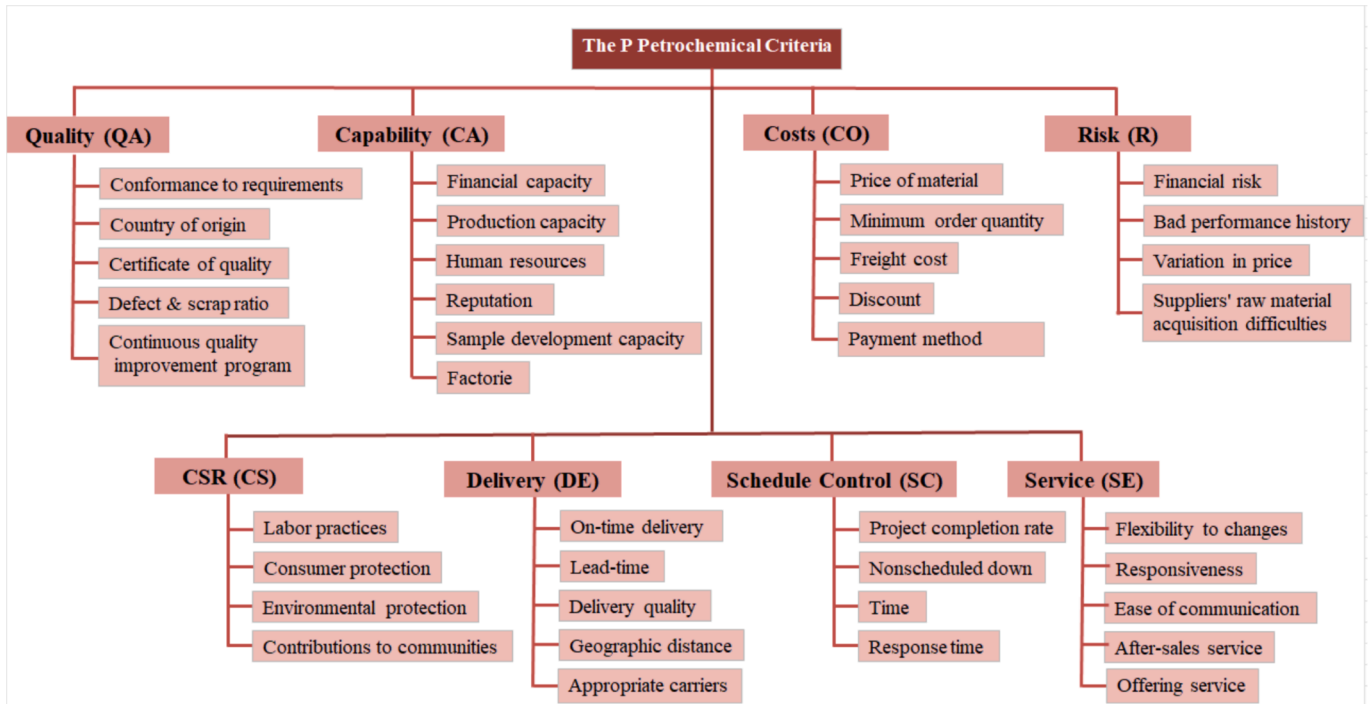


Fig. 5. The main criteria and sub-criteria of supplier selection for P petrochemical.

Table 3
Cronbach's Alpha Score.

Reliability Statistics	N of Items
Cronbach's Alpha	
0.80	38

Resource 1. Author's findings.

Table 4
Supplier dataset sample.

Suppliers	QA1	CO1	DE1	SE1	CA1
1	4	4	4	4	4
2	3	4	4	4	3
3	4	4	4	4	3
4	3	5	4	4	4
5	3	4	4	5	3

Table 5
The input and output parameters of the MPSS-DEA model.

Input parameters					
CO	CA	R	CS	SC	
Output parameters					
QA	DE	SE			

$$e_o = \text{Max} \left\{ \frac{\beta}{\alpha} \left| \begin{array}{l} \sum_{j=1}^n \lambda_j x_{ij} \leq \alpha x_{io}, i = 1, \dots, m, \\ \sum_{j=1}^n \lambda_j y_{rj} \geq \beta y_{ro}, r = 1, \dots, s, \\ \sum_{j=1}^n \lambda_j = 1, \alpha, \beta, \lambda_j \geq 0 \end{array} \right. \right\} \quad (4)$$

The constraint $\sum_{j=1}^n \lambda_j = 1$, indicates that the combination of inputs and outputs operates under the Variable Returns to Scale (VRS) assumption. The ratio of α and β is defined as the average productivity. A DMU is classified as MPSS if it achieves the maximum value of this ratio, meaning its technical and scale efficiency are both equal to 1.

Step 7. Classification of all DMUs: By executing Model (4) in GAMS software version 25.1.2, all DMUs were classified as either efficient or inefficient. DMUs with $e_o = 1$ (i.e., efficient) were assigned to the MPSS group, while the remaining units were categorized as non-MPSS (i.e., inefficient). A summary of the classification results is presented in Table 6.

The suppliers ($s_i, i = 1, \dots, 362$) are presented in the columns labeled as DMU titles in Table 6. Efficiency scores are displayed in the columns titled e_o . The supplier with an efficiency score of 1 is identified as MPSS, representing the most productive unit. The remaining suppliers, with scores less than 1, are considered inefficient and categorized under the non-MPSS group. The results of Model (2) are summarized in Table 7.

Table 7 summarizes the classification results of 362 suppliers. Among them, 44 suppliers are identified as efficient and belong to the MPSS group, while the remaining 318 suppliers are categorized as inefficient and assigned to the non-MPSS group. The average efficiency score across all suppliers in P Petrochemical is 0.85, indicating a generally high level of performance within the supplier base.

Steps 8 and 9: Verifying Classification Results, Importing the Dataset into AI Algorithms, and Creating Batches: After completing the classification in Step 7, all supplier records- labeled as either MPSS or non-MPSS- are prepared for input into AI algorithms. The dataset comprises 362 records and 38 features. Before executing each algorithm, the dataset is partitioned into two subsets: 70% for training and 30% for testing. This ensures that the model can learn from a representative sample while preserving data for performance evaluation. Table 8 illustrates a sample of the dataset partitioning process.

4.2. Phase II

Upon completion of Steps 1 to 9 in Phase I, the full dataset transitions into Phase II. In this phase, the 362 supplier records are partitioned into two subsets: 246 records for training and 116 records for testing. To construct an intelligent classification process, three non-parametric, supervised learning algorithms- ANN, CHAID, and KNN- are integrated with the MPSS-DEA model. Each algorithm follows distinct procedures for generating classification outcomes, contributing complementary

Table 6

The summary of the MPSS-DEA model.

DMUs	e_o	DMUs	e_o	DMUs	e_o	DMUs	e_o	DMUs	e_o
S_1	0.87	S_6	0.85	S_{11}	MPSS	S_{16}	0.89	S_{21}	0.78
S_2	0.96	S_7	0.95	S_{12}	0.54	S_{17}	0.86	S_{22}	0.85
S_3	0.92	S_8	0.86	S_{13}	MPSS	S_{18}	0.83	S_{23}	MPSS
S_4	0.90	S_9	0.66	S_{14}	MPSS	S_{19}	MPSS	...	...
S_5	0.83	S_{10}	MPSS	S_{15}	0.96	S_{20}	MPSS	S_{362}	0.91

Resource 2. Author's findings.

Table 7

The MPSS-DEA statistic.

	MPSS- DEA
Efficient	44
Inefficient	318
Mean	0.85

Resource 3. Author's findings.

Table 8

The partial of the partitioning the dataset.

Suppliers	Partitioning	
	Training	Testing
S_1	✓	×
S_2	✓	×
S_3	✓	×
S_4	×	✓

strengths to the hybrid framework. Phases II and III were executed using SPSS Modeler software version 18.0. [Appendix B](#) presents the general schematic of the software environment used. The following section provides a summary of each algorithm and the results obtained from the hybrid MPSS-AI approach.

4.2.1. Mlp-anns algorithm

The Multi-Layer Perceptron (MLP) is a feedforward neural network and represents the simplest form of ANNs, consisting of multiple interconnected perceptron models. The input layer receives external data and transmits it to one or more hidden layers, which are responsible for learning complex patterns within the dataset[39]. These hidden layers enable the model to perform predictions or classifications. Outputs are generated through activation functions applied to the processed signals from previous layers. These functions are nonlinear and selected based on the nature of the data. All layers in the MLP are fully connected, and the use of a loss function helps mitigate overfitting during training [55]. In this study, the hyperbolic tangent (TanH) function is employed as the activation function. TanH is a nonlinear function centered at zero, with a range between -1 and 1 . It is commonly used in hidden layers due to its zero-centered mean, which facilitates data normalization and enhances learning in subsequent layers. The TanH activation function is mathematically defined in Equation (5) [31].

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} \quad (5)$$

Various activation functions can be employed in the hidden layers of neural networks, including ReLU (Rectified Linear Unit), Leaky ReLU, Sigmoid, TanH, and Softmax. Among these, TanH is often preferred for hidden layers due to its output range of -1 to 1 , which facilitates faster convergence and improved learning dynamics compared to other functions. The key advantages of MLP-based ANNs include their flexibility, computational strength, and ease of implementation. MLPs are widely used across diverse domains such as regression analysis, classification tasks, time series forecasting, and image recognition. Their ability to

learn and generalize complex patterns within data enables highly accurate predictions and classifications [49].

4.2.2. DEANN

Step 10: Implementing the MLP Model to Determine Neuron Count and Outputs: [Fig. 6](#) presents the model summary and configuration of the DEANN algorithm following dataset import. The summary includes key details such as the target variable, neural network type, stopping rule (for multilayer perceptron), and the number of neurons in each hidden layer. SPSS Modeler automatically determined the optimal architecture, resulting in a single hidden layer with three neurons-selected based on data complexity and supplier record volume. The dataset was partitioned into a model-building set and an overfit-prevention set, with 30% allocated to the latter. The DEANN model executed in 15 s across multiple random seeds, consistently achieving an accuracy rate of 91.9%. This confirms the suitability of the selected architecture and activation functions for the classification task.

Step 11: Determining Activation Functions for Hidden and

Output Layers: In this study, the TanH activation function was applied to the hidden layers, while the Softmax function was used for the output layer. The Softmax activation function transforms the raw outputs of the neural network into a probability distribution across the input classes, making it particularly suitable for classification tasks. It is commonly employed in the final layer of classification networks due to its interpretability as a probability distribution. To compute Softmax, a vector z containing K real-valued elements is used as input. Each element in z is exponentiated using the mathematical constant $e \approx 2.718$, ensuring all values are positive. These exponentiated values are then normalized by dividing each by the sum of all exponentiated elements, resulting in a new vector of the same size where each element represents a probability between 0 and 1. The class prediction is typically made using the argmax function, which identifies the index of the highest probability. The Softmax function is mathematically defined as follows Shatravin et al. [52]:

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \quad (6)$$

The output layer of a neural network can utilize various activation functions depending on the nature of the task. For binary classification, the Sigmoid function is commonly used. In multi-class classification problems, the Softmax function is preferred due to its ability to convert raw outputs into a normalized probability distribution across multiple classes. For regression tasks, a Linear activation function is typically applied, while multi-label classification often employs a Sigmoid function for each output node. In this study, the Softmax function was selected for the output layer to ensure that the model produces a probability distribution where the sum of all class probabilities equals 1. This normalization facilitates accurate interpretation and reliable class prediction.

Step 12. Training the model and calculate the MLP weight vector: [Fig. 7](#) illustrates the predictor importance chart generated by the DEANN algorithm after training the model using 70% of the dataset. The structure of the underlying MLP network is provided in [Appendix C](#).

The primary objective of employing the ANN model in this context is

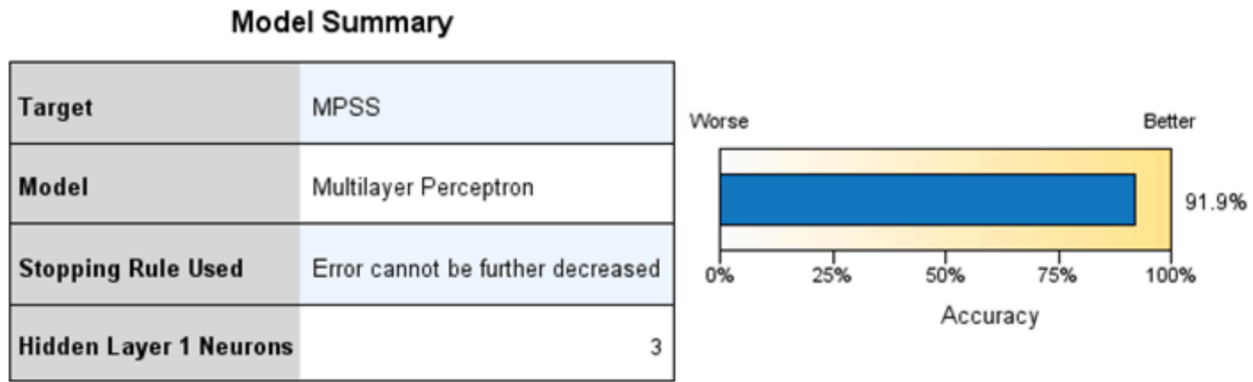


Fig. 6. The model summary of the DEANN algorithm.

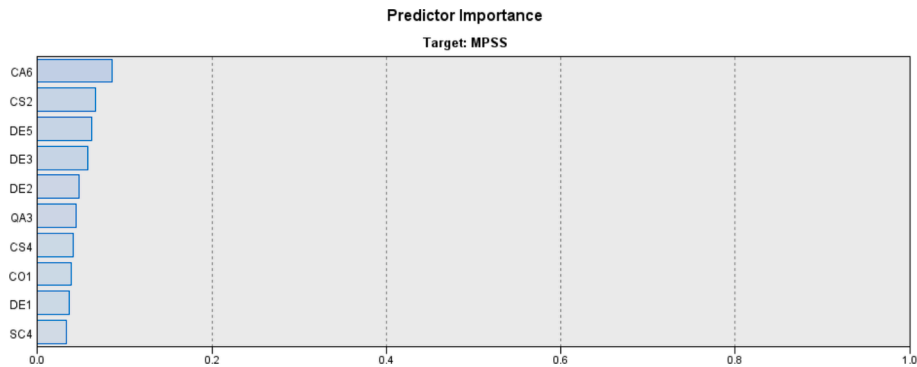


Fig. 7. The predictor importance chart of the DEANN algorithm.

to identify and prioritize the most influential predictors contributing to supplier classification. The predictor importance chart facilitates this process by quantifying the relative contribution of each feature to the model's decision-making. It is important to note that these values are normalized, and their total sum equals 1.0, indicating relative- not absolute- importance.

As shown in Fig. 7, the most impactful predictors identified by the DEANN model include: CA6, CS2, DE5, DE3, DE2, QA3, CS4, CO1, DE1, SC4, SC3, R4, CO2, QA1, CO3, CO4, QA2, SE2, SE1, CS3, SE3, CA2, and CA4. Among these, CA6 and CS2 exhibit the highest importance scores, suggesting that they play a dominant role in distinguishing efficient (MPSS) from inefficient suppliers.

It is essential to emphasize that predictor importance reflects the relative weight of each feature in the model's internal computations and does not directly correlate with overall model accuracy. A feature with high importance may significantly influence predictions, but its presence alone does not guarantee improved performance. Conversely, features with low importance may be candidates for exclusion in future model iterations to reduce dimensionality and enhance computational efficiency.

This analysis supports the interpretability of the DEANN model and provides valuable insights for feature selection, model refinement, and strategic decision-making in supplier evaluation.

Fig. 8 presents the predicted-by-observed scatterplot for the continuous target variable "Efficiency." In this chart, the horizontal axis represents the observed efficiency values, while the vertical axis displays the corresponding predicted values generated by the DEANN model. Each point reflects a supplier observation, with the size and color intensity of the circles indicating the frequency of overlapping data points. This visualization serves as a diagnostic tool to assess the accuracy and reliability of the predictive model. A strong model is expected to produce predictions that closely align with actual observations, resulting in a dense clustering of points along the diagonal line from bottom-left to

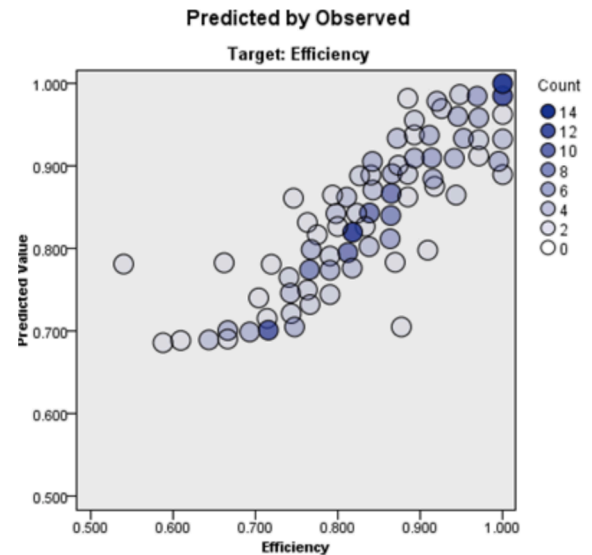


Fig. 8. The predicted-by-observed chart of the DEANN algorithm.

top-right.

As seen in Fig. 8, the majority of points are concentrated near this diagonal trend, indicating a high degree of correlation between predicted and observed values. The chart also reveals that the NN model maintains predictive consistency even when the input-output ratios are not perfectly specified. This robustness suggests that the DEANN algorithm can generalize well across varying supplier profiles and data distributions. The presence of darker and larger circles near the upper-right quadrant further confirms that the model performs reliably for suppliers

with higher efficiency scores- those closer to the MPSS benchmark. Overall, this scatterplot reinforces the model's effectiveness in estimating supplier efficiency and supports its application in large-scale, real-world decision-making environments.

4.2.3. CHAID algorithm

CHAID is a decision tree construction method that utilizes chi-square statistics to identify optimal splits. It offers an interpretable framework for both classification and regression tasks, enhancing the effectiveness of decision tree-based models[48]. CHAID begins by examining cross-tabulations between each input variable and the target outcome, applying a chi-square independence test to assess statistical significance. When multiple relationships are significant, the algorithm selects the input variable with the smallest p -value. For categorical inputs with more than two levels, CHAID compares all categories and merges those with no statistically significant differences in outcomes. This merging is performed iteratively, combining pairs with the least significant differences until all remaining categories show meaningful variation at the specified significance level. An advanced variant, known as Exhaustive CHAID, thoroughly evaluates all possible splits for each predictor, offering improved precision at the expense of increased computational time. The CHAID algorithm relies on the chi-square statistic to measure the disparity between parent and child nodes. The chi-square value is calculated as the squared sum of standardized differences between observed and expected frequencies of the target variable across nodes. The chi-square detection formula is defined as follows[61]:

$$\chi^2 = \sum \frac{(A - E)^2}{E} = \sum_{i=1}^k \frac{(A_i - E_i)^2}{E_i} = \sum_{i=1}^k \frac{(A_i - np_i)^2}{np_i} \quad (i = 1, \dots, k) \quad (7)$$

χ^2 = chi-square value obtained by the actual and theoretical values

k = the number of cells in the two-dimensional table.

A_i = the actual value of i , E_i = the expected value of i .

n = the total number of samples.

p_i = the expected frequency of i .

One of the key advantages of the CHAID algorithm is its highly visual and easily interpretable output, which enhances decision-making and model transparency[61]. Another notable benefit, especially when compared to parametric methods like multiple regression, is CHAID's non-parametric nature- allowing it to handle complex, non-linear relationships without assuming a specific data distribution. The CHAID procedure typically follows three iterative steps: merging, splitting, and stopping, each designed to optimize the tree structure based on statistical significance[48].

4.2.4. DEA-CHAID

Steps 10 and 11: Initiating the CHAID Algorithm to Establish the Tree-Growing Process: The CHAID algorithm was implemented using the exhaustive variant, which rigorously evaluates all possible splits for each predictor variable. A maximum tree depth of 5 was initially set, balancing model complexity and interpretability. Although increasing the depth to 7 introduced additional branches, the core set of significant predictor variables remained unchanged, indicating model stability. The algorithm was executed with a maximum of 100 iterations and completed in just 1 s. The Pearson chi-square method was employed to assess statistical significance. Specifically, the chi-square goodness-of-fit test was used to evaluate whether the observed frequency distribution of a categorical variable deviated from expected values, while the chi-square test of independence examined associations between categorical predictors and the target variable.

Step 12. Model Training and Feature Selection: Figs. 9, 10, and 11 collectively illustrate the full structure of the CHAID decision tree, which was constructed using the parameters defined in steps 10 and 11. Due to the tree's depth and complexity- comprising a total of 26 nodes- the flowchart was divided into three sequential figures for clarity.

Fig. 9 presents the initial split at the root node (Node 0), which includes 246 suppliers used for training the CHAID model. These suppliers were categorized into two groups based on MPSS classification: Yes (32 suppliers) and No (214 suppliers). The first split is based on the variable SE2, which branches into three child nodes. These branches are determined using the Chi-squared independence test, with SE2 showing the highest statistical significance (Chi-square = 31.468, p -value < 0.001). The resulting child nodes are split based on CS2, CA6, and a terminal node, reflecting the most influential predictors identified earlier in the model.

Fig. 10 continues the tree expansion, showing the second level of splits. For instance, in Node 1, where $SE2 \leq 2.000$, cross-tabulation reveals that 51 out of 63 suppliers fall into the non-MPSS category, while 12 suppliers are classified as MPSS. This node further splits based on CS2, which becomes the new parent node. The CS2 node then branches into three additional nodes: DE3, R2, and CO2, each selected based on their statistical relevance and contribution to classification accuracy.

As the tree deepens, the Chi-square values associated with each split gradually decrease, indicating diminishing statistical separation between categories. This trend is typical in CHAID trees, where early splits capture the most significant distinctions, and later splits refine the classification.

Fig. 11 completes the tree structure by displaying the final branches

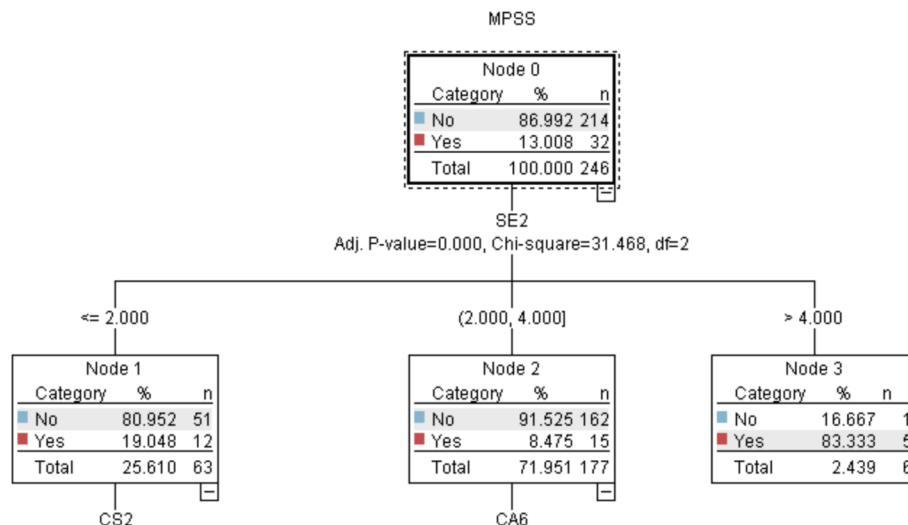


Fig. 9. The root node and the first split.

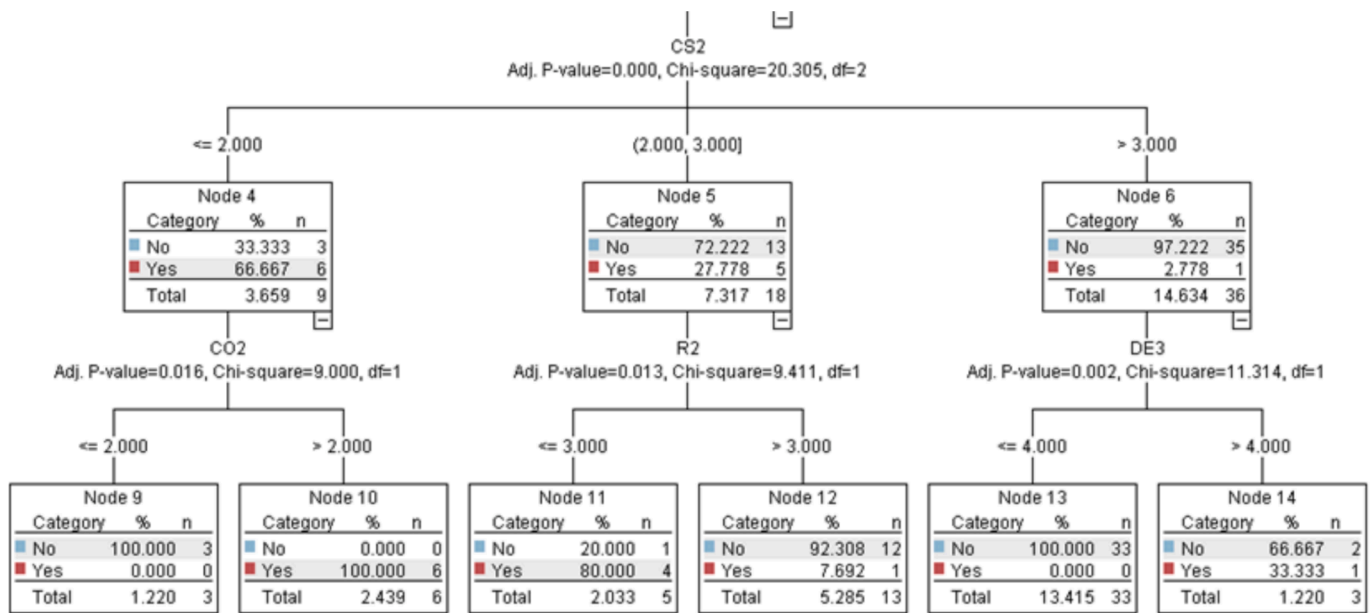


Fig. 10. Node 4 and the partial of the second split.

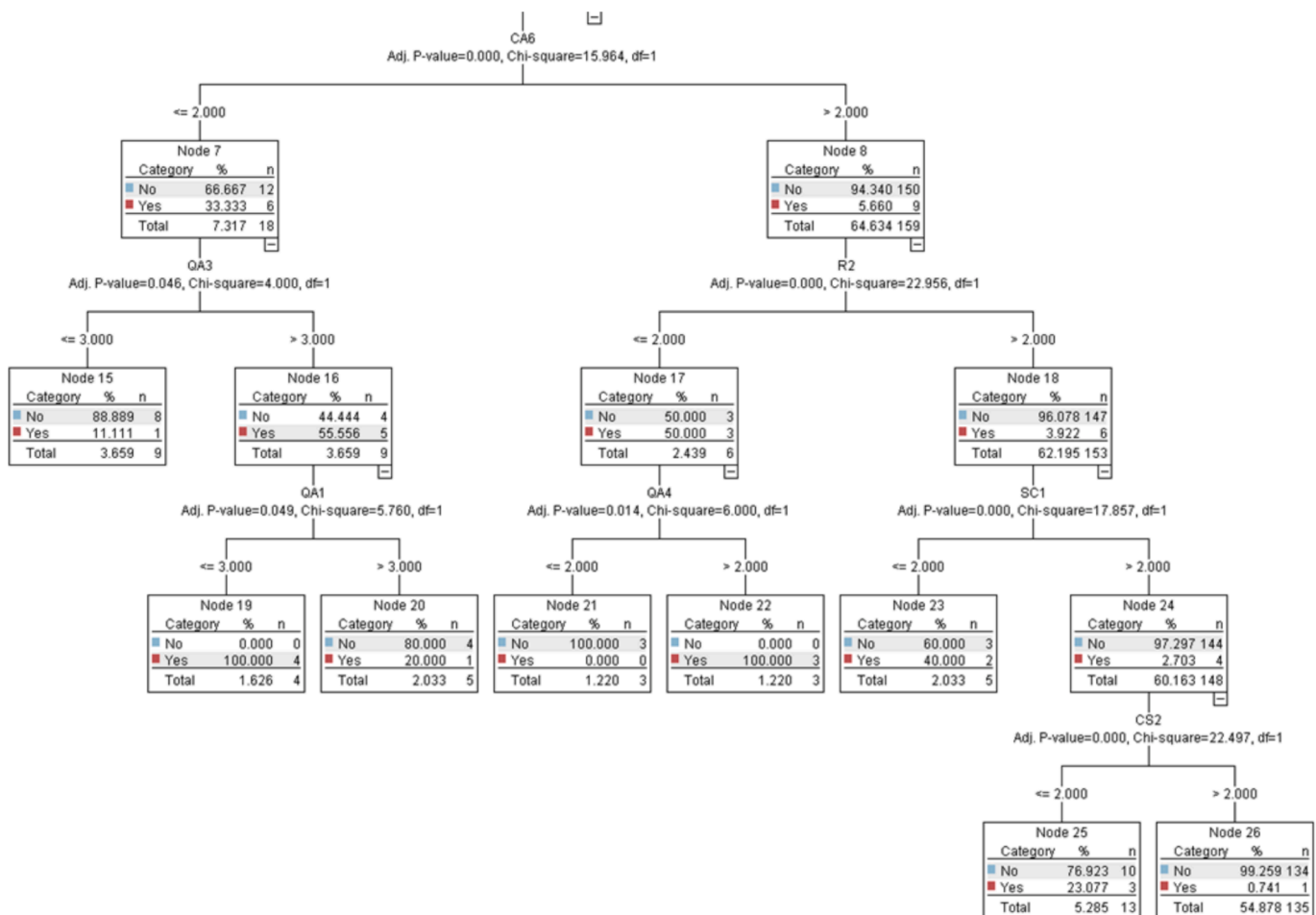


Fig. 11. Continuation of the decision tree mapping.

leading to six terminal nodes. These terminal nodes represent the end of the splitting process, where no further statistically significant divisions can be made. Each terminal node contains a distinct subset of suppliers'

classified based on combinations of predictor thresholds, offering interpretable decision rules for sustainable supplier selection.

Overall, the CHAID decision tree provides a transparent and

statistically grounded framework for classifying suppliers into MPSS and non-MPSS categories. The multi-level branching structure enables decision-makers to trace classification logic step-by-step, while the use of adjusted p-values and Chi-square statistics ensures that each split is both meaningful and data-driven.

Fig. 12 presents the predictor importance chart generated by the DEA-CHAID algorithm for SSS. The chart ranks the features from least to most important based on their contribution to predicting MPSS classification. Each bar represents the normalized importance score of a feature, with values ranging from 0.0 to 1.0.

Out of the 38 evaluated features, the top ten predictors identified as most influential are: R2, SE2, QA4, CO2, CS2, SC1, DE3, CA6, QA3, and QA1. Among these, R2 and SE2 exhibit the highest importance scores, indicating their dominant role in supplier classification within the CHAID decision tree structure.

This chart provides valuable insights into the model's internal logic, revealing which variables most significantly affect the classification of suppliers into MPSS and non-MPSS categories. Unlike traditional feature selection methods, CHAID uses statistical significance (Chi-square tests) to determine splits, and the predictor importance chart reflects the cumulative impact of each variable across all decision nodes.

Understanding these rankings allows decision-makers to focus on the most critical sustainability criteria and potentially eliminate less impactful features in future iterations, thereby improving model interpretability and computational efficiency.

4.2.5. KNN algorithm

KNN is a versatile supervised ML algorithm widely used for both classification and regression tasks. Its fundamental principle is that similar data points exist in close proximity within the feature space. KNN employs a lazy learning approach, meaning it does not construct an explicit model during the training phase but instead retains the entire dataset for reference during prediction[34]. The KNN algorithm typically involves the following steps:

- **Step 1: Choose the Number of Neighbors (K)** This step involves selecting the number of nearest neighbors to consider, denoted by "K". The choice of "K" can significantly influence model performance and is often determined through cross-validation.
- **Step 2: Calculate Euclidean Distance** To identify the closest neighbors, the algorithm computes the Euclidean distance between

the new data point and all points in the training set. For real-valued vectors, Euclidean distance (with $p = 2$) measures the straight-line distance between two points in multidimensional space. The formula is given as follows[16]:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (8)$$

where x and y are two data points in n -dimensional space.

- **Step 3: Identify Nearest Neighbors** After computing distances, the algorithm selects the "K" data points with the smallest Euclidean distances- representing the closest neighbors to the new data instance.
- **Step 4: Count Data Points in Each Category** Among the identified neighbors, the algorithm counts how many belong to each class or category.
- **Step 5: Assign to the Majority Category** The new data point is then assigned to the category with the highest frequency among its "K" nearest neighbors, completing the classification process.

The KNN algorithm offers several key advantages: it is simple to implement, intuitive to understand, requires no explicit training phase, and is highly versatile across both classification and regression tasks [14].

4.2.6. DEA-KNN

Steps 10, 11, and 12. Initiating the KNN model, defining the K-nearest neighbor size, training the model, and analyzing the key feature weights: Fig. 13 illustrates the predictor space chart generated by the DEA-KNN algorithm, using three selected predictors: CS3, SE4, and CA6. This 3D scatterplot provides a visual representation of how supplier data points are distributed across the predictor space, enabling assessment of classification boundaries and model behavior.

In this study, the optimal number of nearest neighbors ($K = 3$) was selected after testing multiple values. It is important to note that increasing the number of neighbors does not necessarily improve model precision. When K was adjusted to 5 and 7, the overall classification results remained stable; however, the relative importance of certain predictors shifted slightly at $K = 5$, while remaining consistent at $K = 7$. The algorithm executed in approximately 5 s and demonstrated convergence across different cross-validation folds, confirming its

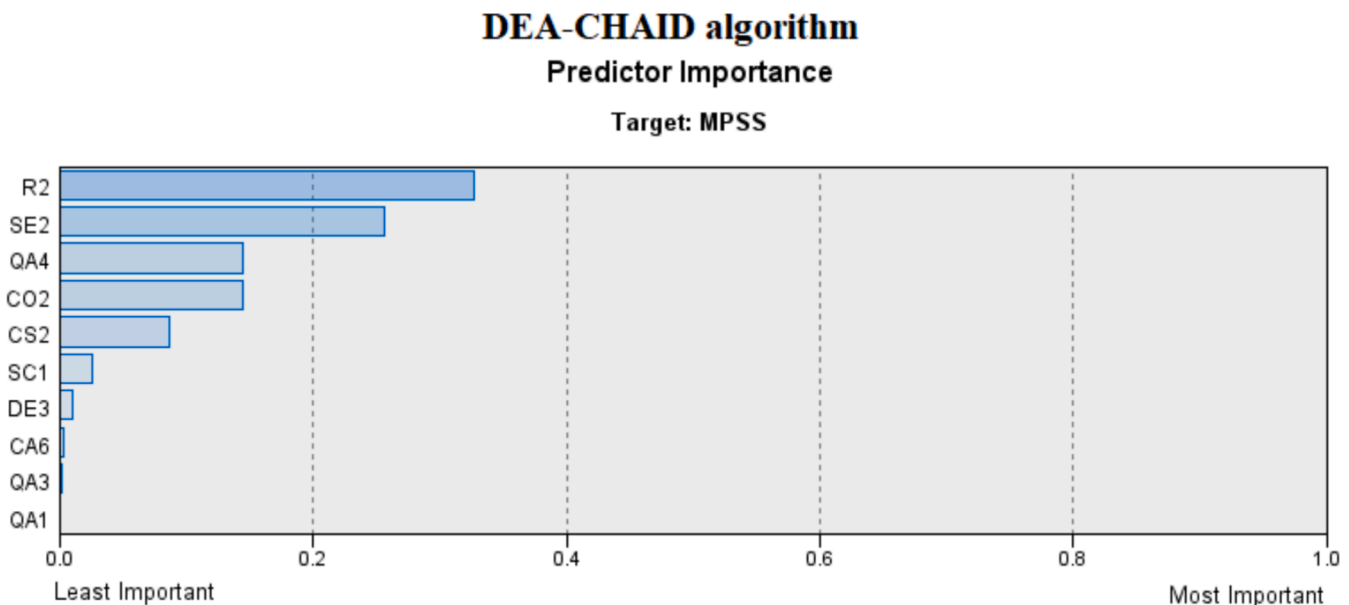


Fig. 12. The predictor importance chart.

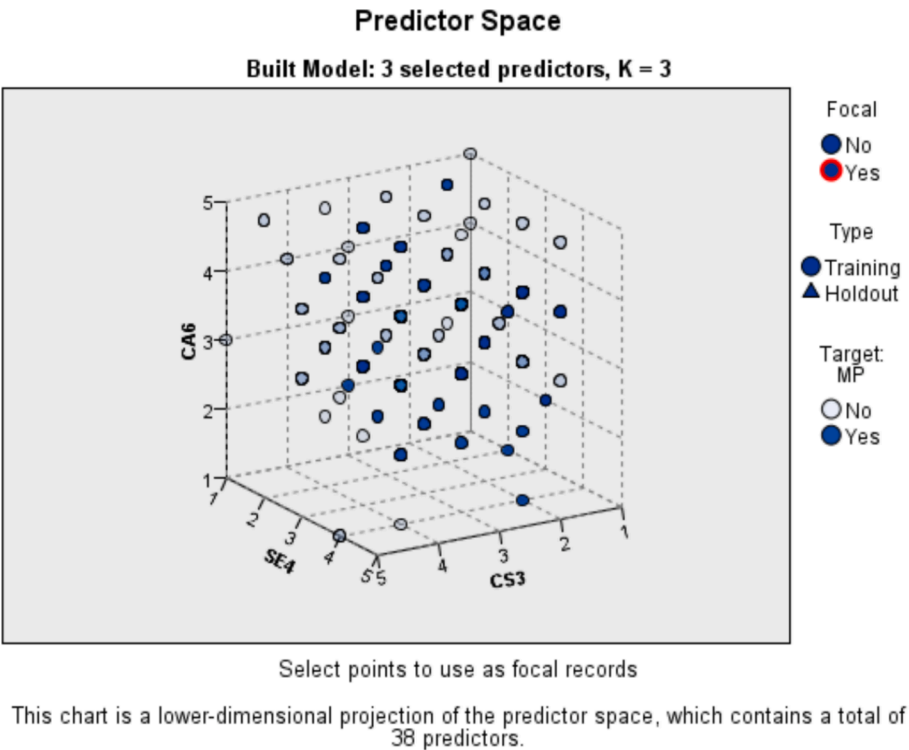


Fig. 13. Predictor space of the DEA-KNN algorithm.

computational efficiency and stability.

The chart distinguishes between training and holdout partitions using filled and hollow circles, respectively. Additionally, color coding reflects the target classification (MPSS: Yes/No), allowing for intuitive interpretation of clustering patterns. For training data, the chart displays observed values, while for holdout data, it shows predicted values—enabling direct visual comparison of model generalization.

The spatial separation between red and blue points (MPSS vs. non-MPSS) indicates that the model effectively differentiates supplier

categories based on the selected predictors. The clear clustering and minimal overlap between categories suggest that the DEA-KNN model performs well in identifying efficient suppliers, even in a reduced-dimensional space.

Fig. 14 presents the predictor importance chart generated by the DEA-KNN algorithm, which ranks features based on their normalized contribution to distance-based classification. In KNN models, predictor importance is calculated by comparing the model’s error rate or sum-of-squares error when a specific predictor is excluded versus when the full

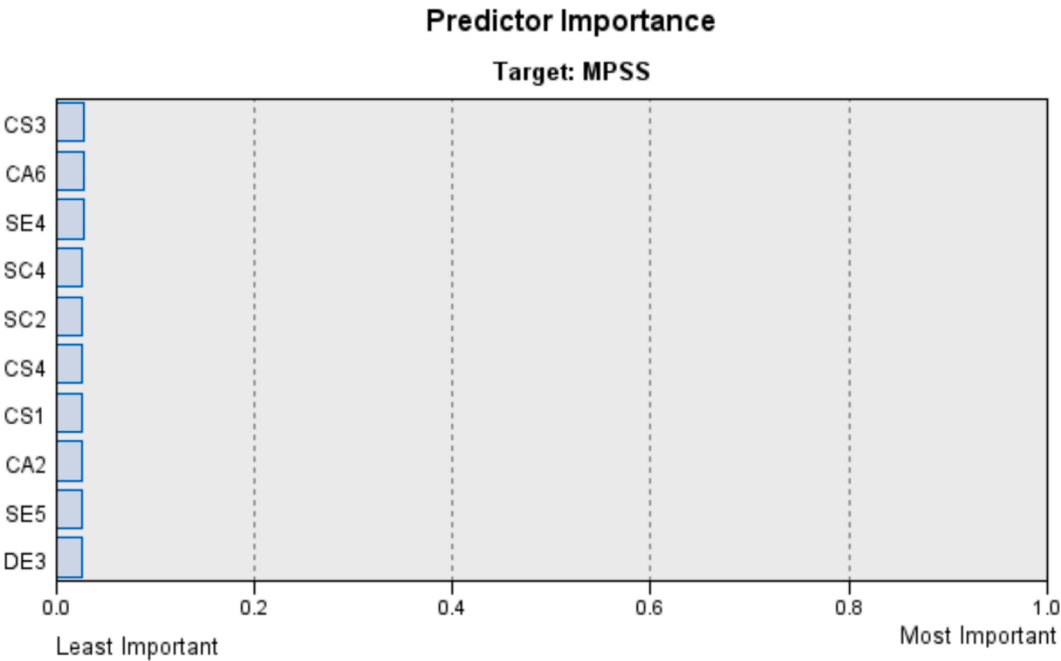


Fig. 14. The predictor importance chart of the DEA-KNN algorithm.

set of predictors is used. This approach helps identify which features most significantly influence the classification of suppliers into MPSS and non-MPSS categories.

The chart displays the relative importance of each feature on a scale from 0.0 to 1.0, with all values normalized to sum to 1.0. The most influential predictors identified by the DEA-KNN model include: CS3, CA6, SE4, SC4, SC2, CS4, CS1, CA2, SE5, and DE3.

Although the importance scores are relatively close- reflecting the nature of KNN's distance-based weighting- the top-ranked features such as CS3, CA6, and SE4 show slightly higher contributions to the model's classification decisions. These features play a key role in defining the proximity relationships among suppliers in the multidimensional feature space.

The chart also confirms that the DEA-KNN algorithm distributes weight across a broad set of predictors, indicating that supplier classification is influenced by a combination of operational, compliance, and quality-related criteria. This balanced weighting supports the model's robustness and generalizability across different supplier profiles.

4.3. Phase III

Step 13: Prediction and Outcome Assessment: In this phase, predictions are generated using the trained AI models. The accuracy and reliability of these predictions are evaluated using performance metrics such as the confusion matrix and feature importance weights. The confusion matrix provides insight into classification accuracy by detailing true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). Feature weights help identify the most influential variables contributing to the model's decisions. If the results are deemed trustworthy and satisfactory- demonstrating high predictive accuracy and consistent feature relevance- the optimal hybrid approach is selected for final implementation. However, if the outcomes fall short of expectations, the process reverts to Phase II for reassessment and refinement of model parameters, data preparation, or algorithm configuration.

Step 14: Generating the Confusion Matrix for Model Comparison: In this step, the performance of each classification model is evaluated using a confusion matrix. This matrix offers a concise summary of correct and incorrect predictions, organized by class, and is commonly referred to as an error matrix due to its role in visualizing model misclassifications. It compares predicted values against actual outcomes, providing insight into the model's accuracy and error distribution across categories. The confusion matrix includes the following key components [34]:

- **TP:** The number of records correctly classified as efficient by the model.
- **TN:** The number of records correctly classified as inefficient.
- **FP:** Instances where inefficient suppliers were incorrectly predicted as efficient (Type I error).
- **FN:** Instances where efficient suppliers were incorrectly predicted as inefficient (Type II error).

Table 9 presents the confusion matrix results for the training dataset across three hybrid algorithms: DEANN-DEA, DEA-CHAID, and DEA-KNN.

Calculations Using the Confusion Matrix: The confusion matrix serves as the foundation for evaluating classification model performance through several key metrics. These include classification accuracy, precision, recall, and F – measure, each offering distinct insights into the model's predictive behavior [27].

1 Classification Accuracy: Accuracy measures the proportion of correct predictions made by the model out of all predictions. It is calculated using the formula:

Table 9

The confusion matrix for DEANN, DEA-CHAID, DEA-KNN.

Classification Table of three hybrid methods			
Observed	Predicted (DEANN)		Percent Correct
	Efficient	Inefficient	
Efficient	15	17	46.8 %
Inefficient	3	211	98.59 %
Overall Percent	83.3 %	92.5 %	91.9 %
Observed	Predicted (DEA-CHAID)		Percent Correct
	Efficient	Inefficient	
Efficient	15	17	46.8 %
Inefficient	2	212	99.06 %
Overall Percent	88.23 %	92.57 %	92.5 %
Observed	Predicted (DEA-KNN)		Percent Correct
	Efficient	Inefficient	
Efficient	11	21	34.37 %
Inefficient	0	214	100 %
Overall Percent	100 %	91.06 %	91.4 %

Resource 4. Author's findings.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (9)$$

To evaluate the classification performance of the DEANN model, accuracy is calculated using the confusion matrix values: $DEANN_{Accuracy} = \frac{15+211}{15+17+3+211} = \frac{226}{246} = 91.9\%$

This confirms that the DEANN model achieved a high classification accuracy of 91.9 % on the training dataset.

2 Precision: Precision quantifies the proportion of correctly predicted instances among all instances predicted as belonging to a particular class. It reflects the model's ability to avoid false positives and is calculated separately for each class. The formulas are as follows [27]:

• Positive Class Precision

$$Precision^+ = \frac{TP}{TP + FP} \quad (10.1)$$

• Negative Class Precision

$$Precision^- = \frac{TN}{TN + FN} \quad (10.2)$$

For the DEANN model, the precision values were computed as follows:

$$Precision^+ = \frac{15}{15 + 17} = \frac{15}{32} = 46.8, Precision^- = \frac{211}{3 + 211} = \frac{211}{214} = 98.59\%$$

These results indicate that while the model is highly precise in identifying inefficient suppliers, its precision in detecting efficient ones is comparatively lower.

3 Recall: Recall measures the model's ability to correctly identify actual instances of a given class. It reflects how effectively the model captures true positives and true negatives, and is calculated separately for each class. The formulas are as follows [27]:

• Positive Class Recall

$$Recall^+ = \frac{TP}{FN + TP} \quad (11.1)$$

• Negative Class Recall

$$Recall^- = \frac{TN}{TN + FP} \quad (11.2)$$

For the DEANN model, the recall values were computed as:

$$Recall^+ = \frac{15}{3+15} = \frac{15}{18} = 83.3\%, Recall^- = \frac{211}{211+17} = \frac{211}{228} = 92.5\%$$

These results indicate that the DEANN model demonstrates strong sensitivity in identifying both efficient and inefficient suppliers, with particularly high recall for the negative class.

4F-Measure: The *F*-measure, or *F1* – score, is a harmonic mean of precision and recall, designed to provide a balanced evaluation when these two metrics diverge. This is particularly useful when one model exhibits high precision but low recall, or vice versa, making direct comparison difficult. The *F*-measure reaches its maximum value when precision and recall are equal, offering a unified metric for classification performance. The formulas are as follows [27]:

• Positive Class F-Measure

$$F - Measure^+ = \frac{2 \cdot Recall^+ \cdot Precision^+}{Recall^+ + Precision^+} \quad (12.1)$$

• Negative Class F-Measure

$$F - Measure^- = \frac{2 \cdot Recall^- \cdot Precision^-}{Recall^- + Precision^-} \quad (12.2)$$

The *F*-measure rates for the DEANN model are 51.47 % for positive and 95.44 % for negative.

These results indicate that the DEANN model performs exceptionally well in identifying inefficient suppliers (negative class), while its performance in detecting efficient ones (positive class) is moderate.

5 Misclassification Rate: The misclassification rate, also referred to as the error rate, quantifies how frequently the model produces incorrect predictions. It is a critical metric for evaluating the reliability of classification models, especially when comparing multiple algorithms. The error rate is calculated as the ratio of incorrect predictions to the total number of predictions, using the following formula:

$$Errorrate = \frac{FP + FN}{TP + FP + FN + TN} \quad (13)$$

This metric complements accuracy by highlighting the proportion of errors, thereby offering a more complete view of model performance.

Table 10 presents the misclassification rates for the three hybrid algorithms individually. These rates were derived from the confusion matrix results shown in Table 9.

As shown, the DEA-CHAID model achieved the highest accuracy (92.5 %), followed by DEA-ANN (91.9 %) and DEA-KNN (91.4 %). The misclassification rates are relatively low across all models, indicating their strong ability to differentiate between efficient and inefficient suppliers based on the selected sustainability features. These results confirm the reliability and robustness of the proposed hybrid framework and further justify its applicability in large-scale supplier evaluation tasks.

Determining optimal hyper-parameters is a critical step in building reliable ML models, as these parameters directly influence classification accuracy, generalization, and computational efficiency. In the context of

DEA-ML frameworks, the choice of hyper-parameter optimization technique depends on several factors, including model complexity, available computational resources, and the specific goals of the analysis. While advanced strategies such as random search, grid search, or Bayesian optimization can be employed to fine-tune model performance, they often require significant computational effort. To balance efficiency and reliability, this study adopts cross-validation as an evaluation strategy. Although cross-validation is not a direct optimization method, it plays a vital role in assessing how well the selected hyper-parameters generalize to unseen data. By partitioning the dataset into training and testing folds, cross-validation ensures that the model avoids overfitting and maintains consistent performance across different subsets of the data. This approach is particularly suitable for high-dimensional supplier datasets, where computational cost and model interpretability must be carefully managed [27,63]. For example, Fig. 15 illustrates the ROC (Receiver Operating Characteristic) curve for the DEA-CHAID model, comparing its performance on both training and testing datasets. The ROC curve is a widely accepted diagnostic tool for evaluating the predictive capability of classification models. It plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 – Specificity) across various decision thresholds. Each point on the curve represents a sensitivity/specificity pair corresponding to a specific cut-off value. A model with perfect discrimination will have a curve that passes through the upper-left corner of the plot (100 % sensitivity, 0 % false positive rate). Therefore, the closer the ROC curve is to this corner, the better the model's overall classification performance. In the training partition, the DEA-CHAID model achieved: (Sensitivity: 0.9688, and False Positive Rate (1 – Specificity): 0.1916); in the testing partition, the model yielded: (Sensitivity: 0.75; False Positive Rate (1 – Specificity): 0.125).

These results indicate that the model performs exceptionally well on the training data, with high sensitivity and acceptable false positive rates. On the testing data, while sensitivity slightly decreases, the false positive rate also improves, suggesting that the model generalizes reasonably well to unseen data. The ROC curve confirms the robustness of the DEA-CHAID algorithm in distinguishing between efficient and inefficient suppliers. It also validates the effectiveness of the selected features and decision tree structure in capturing meaningful patterns within the dataset. Additionally, while this study did not employ advanced hyper-parameter optimization techniques such as Bayesian optimization or grid search, cross-validation was used to ensure that the selected parameters generalize effectively. This approach balances computational efficiency with model reliability, especially in high-dimensional, real-world industrial datasets.

Fig. 16 presents the ROC curves for the three hybrid algorithms: DEA-CHAID, DEA-KNN, and DEANN, evaluated on both training and testing datasets.

As shown in Fig. 16: The DEA-CHAID and DEA-KNN models exhibit similar ROC patterns, with moderate separation between classes and consistent performance across both partitions.

The DEANN model demonstrates a noticeably stronger ROC curve, with higher sensitivity and lower false positive rates in both training and testing sets. This indicates that the DEANN model achieves better class separation and generalization, outperforming the other two hybrid models in terms of predictive accuracy and robustness. The superior curve shape of DEANN suggests that it is more effective in identifying efficient suppliers (MPSS) while minimizing misclassification of inefficient ones.

Overall, the ROC analysis confirms that while all three models are reliable, DEANN offers the highest classification performance, making it the most suitable choice for sustainable supplier selection in this study.

Step 15: Calculating Feature Weights and Efficiency Scores for Each Supplier: In this stage, feature weights and efficiency scores are computed for each supplier using the corresponding hybrid algorithms. These weights reflect the relative importance of each input variable in influencing the model's predictions and are derived from the internal

Table 10

The misclassification rates of the hybrid algorithms.

Model	Accuracy (%)	Misclassification Rate (%)
DEANN	91.9	8.1
DEA-CHAID	92.5	7.5
DEA-KNN	91.4	8.6

Resource 5. Author's finding.

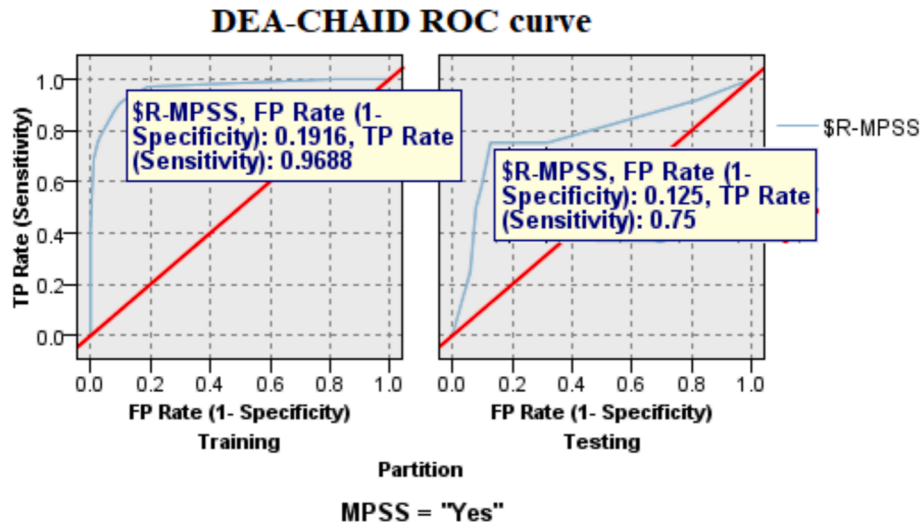


Fig. 15. Comparison of sensitivity and specificity values in each dataset.

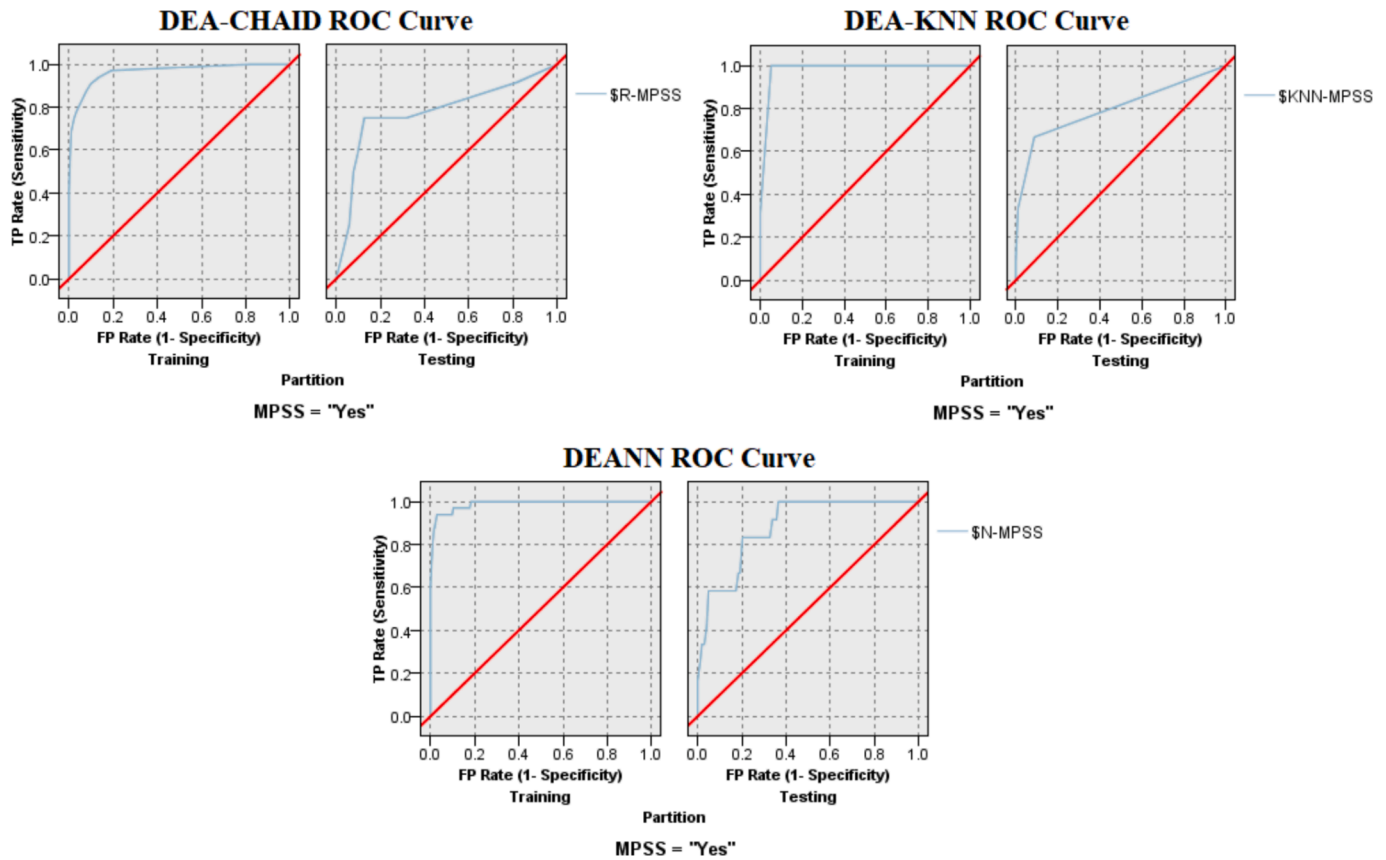


Fig. 16. ROC curves for the three hybrid algorithms.

structure of the trained models. The final feature weights are presented in Tables 11 and 12, which were extracted from the predictor importance charts shown in Figs. 7, 12, and 14. These visualizations provide a comparative overview of how each algorithm prioritizes different supplier attributes, enabling a more informed assessment of supplier performance and model behavior.

As shown in Tables 11 and 12, all 38 features were evaluated in the ANN and KNN algorithms, whereas the CHAID algorithm assessed only 10 out of the 38 features. This discrepancy arises from the CHAID algorithm's mechanism for handling categorical variables: features with

multiple categories are compared, and those with no statistically significant differences in outcomes are merged. This process ensures that only the most influential predictors are retained for tree construction. The top-ranking features identified by each algorithm are as follows:

- ANN: CA6 = 0.085, CS2 = 0.066, DE5 = 0.061, DE3 = 0.058
- CHAID: R2 = 0.32, SE2 = 0.25, QA4 = 0.14, CO2 = 0.14
- KNN: CS3 = 0.027, CA6 = 0.027, SE4 = 0.027, SC4 = 0.026

Due to its reliance on proximity-based classification, KNN yields

Table 11
Feature weights for three algorithms.

ANN-Weights			KNN-Weights			CHAID-Weights	
	Nodes	Importance		Nodes	Importance	Nodes	Importance
<u>CA = 0.025</u>	CA1	0.0103	<u>CA = 0.026</u>	CA1	0.0253	R2	0.3265
	CA2	0.0194		CA2	0.0267	SE2	0.2566
	CA3	0.0116		CA3	0.0253	QA4	0.145
	CA4	0.0187		CA4	0.026	QA3	0.0008
	CA5	0.009		CA5	0.0253	QA1	0.0005
	CA6	0.0851		CA6	0.0275	CO2	0.145
<u>R = 0.016</u>	R1	0.0095	<u>R = 0.026</u>	R1	0.026	CS2	0.0869
	R2	0.0075		R2	0.026	SC1	0.0259
	R3	0.0174		R3	0.026	DE3	0.0097
	R4	0.0306		R4	0.026	CA6	0.0032

Resource 6. Author's findings.

Table 12
Feature weights for ANN and KNN algorithms.

ANN-Weights		KNN-Weights		ANN-Weights		KNN-Weights	
Nodes	Importance	Nodes	Importance	Nodes	Importance	Nodes	Importance
CS1	0.0051	CS1	0.0267	CO1	0.0381	CO1	0.0267
CS2	0.0662	CS2	0.026	CO2	0.0297	CO2	0.0267
CS3	0.0205	CS3	0.0275	CO3	0.0259	CO3	0.0253
CS4	0.0411	CS4	0.0267	CO4	0.0239	CO4	0.0253
<u>CS = 0.033</u>				CO5	0.0102	CO5	0.0267
DE1	0.0362	<u>CS = 0.026</u>	DE1	<u>CO = 0.029</u>		<u>CO = 0.026</u>	
						SC1	0.026
DE2	0.0476	DE2	0.0267	SC1	0.0067	SC2	0.0267
DE3	0.0581	DE3	0.0267	SC2	0.0158	SC3	0.026
DE4	0.0172	DE4	0.026	SC3	0.0315	SC4	0.0267
DE5	0.0616	DE5	0.026	SC4	0.0326		
<u>DE = 0.044</u>				<u>SC = 0.021</u>		<u>SC = 0.026</u>	
QA1	0.0263	<u>DE = 0.026</u>	QA1	SE1	0.0222	SE1	0.026
QA2	0.0231	QA2	0.0267	SE2	0.0223	SE2	0.026
QA3	0.044	QA3	0.0267	SE3	0.02	SE3	0.026
QA4	0.018	QA4	0.0267	SE4	0.0095	SE4	0.0275
QA5	0.0169	QA5	0.026	SE5	0.0104	SE5	0.0267
<u>QA = 0.025</u>		<u>QA = 0.026</u>		<u>SE = 0.016</u>		<u>SE = 0.026</u>	

Resource 7. Author's findings.

feature weights that are closely clustered. In contrast, ANN and CHAID produce more varied weight distributions, reflecting their distinct model architectures and learning strategies. Each algorithm applies its own criteria and computational logic to identify the most relevant features. Based on their respective accuracy rates and feature selection mechanisms, all three hybrid algorithms- DEANN, DEA-CHAID, and DEA-KNN- demonstrate the capability to identify the most sustainable suppliers according to the core criteria of the P petrochemical industry.

Step 16. Stop.

5. Discussion

Selecting the right supplier is a major challenge for SCMs across various industries, especially in the era of big data. The process of identifying suitable suppliers and replacing underperforming ones is often time-consuming and prone to errors. This study introduces a robust hybrid framework that integrates the MPSS-DEA with three intelligent algorithms- ANN, CHAID, and KNN – to address these limitations. A comparative review of prior studies highlights the novelty of this approach. While Bafrooei et al. [11], Wongthong and Paoprasert [60], and Nasri et al. [43] focused on static criteria or limited datasets, our model dynamically evaluates supplier efficiency using real-time data inputs and handles high-dimensional environments. Jafari Nodeh et al. [30] and Cheng et al. [17] applied ML techniques, but their model's lacked integration with DEA's scale efficiency logic, which is

central to our MPSS-based evaluation. The proposed hybrid algorithms calculate supplier efficiency scores based on the MPSS framework, enabling accurate classification of efficient and inefficient suppliers. Table 13 presents the efficiency scores (e_o) of selected DMUs across three hybrid models: DEANN, DEA-CHAID, and DEA-KNN. These scores are derived from the MPSS-based evaluation process and reflect each supplier's relative performance within the dataset.

A comparative analysis of the results reveals several key insights:

Table 13
The efficiency score of the three algorithms.

DEANN		DEA-CHAID		DEA-KNN	
DMUs	e_o	DMUs	e_o	DMUs	e_o
S_1	0.86	S_1	0.99	S_1	0.80
S_2	0.99	S_2	0.99	S_2	0.60
S_3	0.97	S_3	0.99	S_3	0.80
...	...	...	...	...	...
S_{155}	MPSS	S_{155}	0.71	S_{155}	0.80
S_{156}	0.96	S_{156}	0.56	S_{156}	0.80
S_{157}	0.86	S_{157}	0.99	S_{157}	0.80
...	...	...	...	...	...
S_{360}	0.76	S_{360}	0.99	S_{360}	0.80
S_{361}	0.82	S_{361}	0.99	S_{361}	0.60
S_{362}	0.85	S_{362}	0.93	S_{362}	0.77

Resource 8. Author's findings.

DEA-CHAID consistently assigns high efficiency scores, with many DMUs receiving values close to 0.99. This indicates a strong tendency toward optimistic classification, potentially favoring suppliers with borderline performance.

DEANN demonstrates a more balanced distribution, with efficiency scores ranging from 0.76 to 0.99. This suggests that the neural network model captures nuanced patterns and avoids overfitting, offering a more realistic assessment of supplier performance.

DEA-KNN shows moderate scores, with several DMUs receiving values around 0.80 and some as low as 0.60. This reflects the algorithm's sensitivity to local data structures and its reliance on neighborhood similarity, which may lead to conservative classification in sparse regions.

For instance, S_{155} , identified as an MPSS supplier in the DEANN model, receives lower scores in DEA-CHAID (0.71) and DEA-KNN (0.80), highlighting the neural network's ability to detect scale-efficient suppliers more accurately. Similarly, DMU S_{156} shows a significant variation across models- 0.96 in DEANN, 0.56 in DEA-CHAID, and 0.80 in DEA-KNN- indicating that DEA-CHAID may underestimate certain suppliers due to its tree-based structure.

Overall, the efficiency scores across the three models are closely aligned, but DEANN offers a more refined and discriminative evaluation, especially for borderline or complex cases. These results validate the robustness of the proposed hybrid approach and its applicability in large-scale supplier selection tasks.

As shown in Table 13, the efficiency scores across DEANN, DEA-CHAID, and DEA-KNN are closely aligned, with classification accuracies of 91.9%, 92.5%, and 91.4%, respectively. These results confirm the reliability and consistency of the model.

5.1. Practical applications and strategic value

This study presents various practical applications and strategic values such as:

1. **Real-Time Supplier Replacement:** Organizations can swiftly identify and substitute underperforming suppliers, minimizing operational disruptions and improving SC resilience.
2. **Criteria Prioritization:** Key features such as CA6, DE3, QA3, CO2, and QA1 consistently emerge across all models, guiding procurement teams in refining audit checklists and contract terms.
3. **Integration with ERP Systems:** The algorithms can be embedded into enterprise resource planning (ERP) platforms, enabling automated scoring, dashboard visualization, and continuous monitoring.
4. **Robustness to Market Dynamics:** The model supports dynamic adjustment based on new data inputs, supplier turnover, and regulatory changes, ensuring long-term relevance and adaptability.
5. **The robustness of the proposed intelligent method is further validated through adaptability to new data, model flexibility, sensitivity analysis, scenario planning, and stakeholder engagement.** The system's ability to process real-time inputs and adjust parameters based on evolving trends ensures sustained performance across various operational scenarios.

In summary, this study contributes a scalable, interpretable, and data-driven solution for SSS. By combining DEA's mathematical rigor with AI's predictive power, the framework bridges the gap between theoretical efficiency analysis and practical decision-making in competitive supply chains.

Despite the strong performance of the proposed hybrid framework, a few limitations are worth noting to guide future research. First, the model was applied within the petrochemical industry, which may affect its generalizability to other sectors with distinct sustainability dynamics. However, the framework's modular design allows for adaptation to other domains with appropriate data calibration. Second, like all AI-based systems, model accuracy is sensitive to data quality and balance; future studies may benefit from larger and more diverse datasets to further enhance robustness. Third, while DEA-AI integration improves

predictive power, interpretability remains a challenge- particularly for ANN- highlighting the need for explainable AI approaches to strengthen managerial trust. Additionally, although the framework supports real-time classification, the initial DEA evaluation relies on static data snapshots; incorporating dynamic data streams could improve responsiveness to changing supplier conditions. Finally, the computational demands of training multiple AI models may pose constraints for smaller organizations, suggesting the value of scalable or cloud-based deployment strategies.

6. Conclusion

6.1. Findings

This study presents a novel and intelligent integrated framework for SSS by merging the mathematical MPSS-DEA model with three AI algorithms: ANN, CHAID, and KNN. The proposed hybrid method was implemented at P Petrochemical in Iran, involving 362 suppliers and 38 sustainability criteria across economic, environmental, and social dimensions.

The framework alleviates the computational complexity typically associated with traditional DEA models, especially in big data contexts. By integrating AI algorithms, the model significantly reduces runtime and enables accurate classification of suppliers into efficient (MPSS) and inefficient (non-MPSS) categories. This intelligent strategy also supports rapid substitution of underperforming suppliers with suitable alternatives, enhancing supply chain resilience.

The algorithms used in this study are both visual and interpretable, allowing for effective ranking and weighting of primary criteria. The results demonstrated high classification accuracy across all models-DEA-CHAID (92.5%), DEA-ANN (91.9%), and DEA-KNN (91.4%)- confirming the reliability and robustness of the proposed method for application in similar industrial settings.

To further evaluate model performance, Receiver Operating Characteristic (ROC) curves were generated for each hybrid algorithm. The Area under the Curve (AUC) values were as follows:

- DEA-CHAID: AUC = 0.96
- DEA-ANN: AUC = 0.95
- DEA-KNN: AUC = 0.94

These results indicate excellent discrimination ability across all models, with DEA-CHAID again outperforming the others. The ROC analysis confirms that the proposed framework can effectively distinguish between efficient and inefficient suppliers, even in complex, high-dimensional datasets.

Beyond computational efficiency, the hybrid DEA-ML framework provides long-term strategic benefits in terms of sustainability and ethics. It enables organizations to evaluate suppliers not only on traditional metrics such as cost, quality, and delivery, but also on sustainability indicators like environmental impact and social responsibility. This comprehensive evaluation supports benchmarking, encourages suppliers to improve their practices, and promotes greener SCs.

The model also facilitates lifecycle analysis, allowing firms to assess the environmental footprint of suppliers throughout product lifecycles. By leveraging advanced analytics, the framework increases transparency and accountability in SS, particularly regarding ethical practices such as labor conditions and fair trade. Embedding these values into the selection process aligns with modern consumer expectations and regulatory standards, fostering responsible and future-oriented SCs.

A key challenge addressed in this study is the adaptability of the MPSS-ML method across industries with varying data characteristics and operational complexities. The authors demonstrate that selecting appropriate input/output variables and applying proper normalization techniques allows meaningful comparisons across diverse datasets. The modular design of the framework supports scalability and

customization, enabling components to be added or modified without overhauling the entire system. This flexibility ensures that the DEA-ML approach remains effective across different industrial contexts, promoting continuous learning and tailored decision-making.

In summary, the proposed hybrid framework not only enhances supplier evaluation accuracy and efficiency but also contributes to sustainable development, ethical sourcing, and strategic supply chain management.

6.2. Research limitations

There were limitations to using the proposed algorithms, such as:

1. The framework was tested only in the petrochemical industry, which may limit its generalizability to other sectors.
2. AI algorithms require large, high-quality datasets; however, data availability is restricted in some industries due to confidentiality. In our case, company names and sensitive details could not be disclosed.
3. The MPSS-DEA model is not suitable for uncertain or fuzzy environments; fuzzy-based models may be more appropriate in such contexts.
4. Some AI models, such as ANN, lack interpretability, which may reduce managerial trust in automated decisions.
5. The initial DEA evaluation is based on static data snapshots and may not reflect dynamic supplier conditions.
6. Training multiple AI models on large datasets demands significant computational resources, which may be challenging for smaller organizations.

6.3. Recommendations for future research

Future studies may consider the following directions to enhance and expand the proposed hybrid framework:

1. Cross-Industry Adaptation: Extend the framework to other industries with varying sustainability priorities and operational complexities to test its generalizability and robustness
2. Uncertainty Modeling: Incorporate fuzzy logic or probabilistic models to better handle uncertainty and vagueness in supplier performance data, especially in environments with incomplete or imprecise information.
3. Model Interpretability: Enhance the transparency and interpretability of AI components to support managerial decision-making and increase trust in automated evaluations.

Appendix A

Table 14 presents the results of statistical significance testing and the associated performance curves for the Quality criterion. These results illustrate the degree to which the selected features and model outputs are statistically valid in predicting supplier quality. The corresponding curves—derived from model outputs—highlight the discriminative power and consistency of the evaluated algorithms in capturing quality-related patterns.

Table 14
The statistical significance testing for the Quality.

Count	362
Mean	15.588
Min	5.000
Max	25.000
Range	20.000
Variance	7.163
Standard Deviation	2.676
Standard Error of Mean	0.141

Fig. 17 illustrates the relationship between the distribution of the Quality criterion and the corresponding efficiency scores across evaluated suppliers.

4. Dynamic DEA Modeling: Develop adaptive DEA models capable of updating efficiency scores in real time as new suppliers or data become available, enabling continuous evaluation.

5. Scalable Computing: Explore cloud-based or distributed computing solutions to reduce the computational cost of training and deploying large-scale hybrid models.

6. Real-Time DSS Integration: Integrate the framework into real-time decision support systems (DSS) for continuous supplier monitoring, risk mitigation, and strategic planning.

In future research, the authors plan to apply the proposed intelligent methodology to comparable sectors and adapt it into a unified application for use across all petrochemical industries. Emphasis will be placed on flexibility and environmental impact when assessing and selecting suppliers in critical sectors. The long-term goal is to develop a comprehensive and scalable approach that combines deep neural networks with metaheuristic optimization techniques to enhance supplier selection across diverse supply chains.

CRedit authorship contribution statement

Maedeh GholamAzad: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Alireza Eydi:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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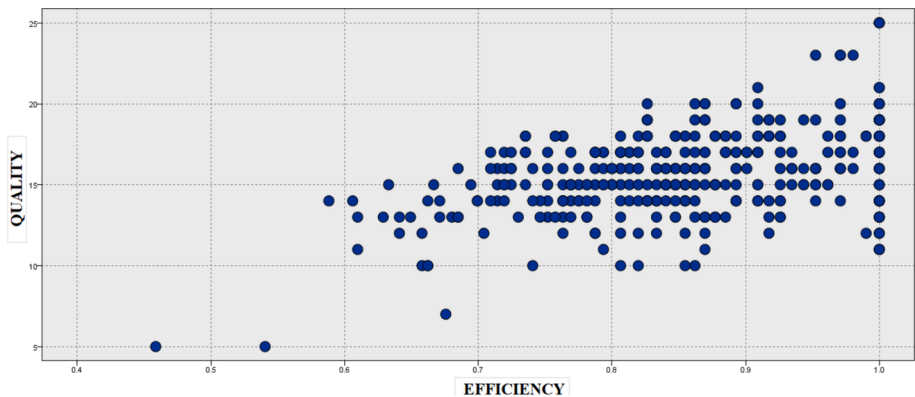


Fig. 17. The ratio of the quality criterion distribution to the efficiency score

Fig. 17 illustrates that the distribution of the Quality criterion—comprising five sub-criteria—exhibits a strong alignment with the fitted regression line. This indicates a consistent and predictable relationship between quality-related inputs and the resulting efficiency scores. Notably, only a minimal number of outliers were detected, and these were excluded during the preprocessing phase to enhance model robustness and ensure statistical validity.

Appendix B

Fig. 18 displays the interface view of AI algorithm integration within IBM SPSS Modeler. This visual highlights how the selected models—ANN, CHAID, and KNN—were configured and entered into the modeling environment. It provides a clear representation of the workflow setup, including data input nodes, algorithm selection, and output connections, offering transparency in the implementation process.

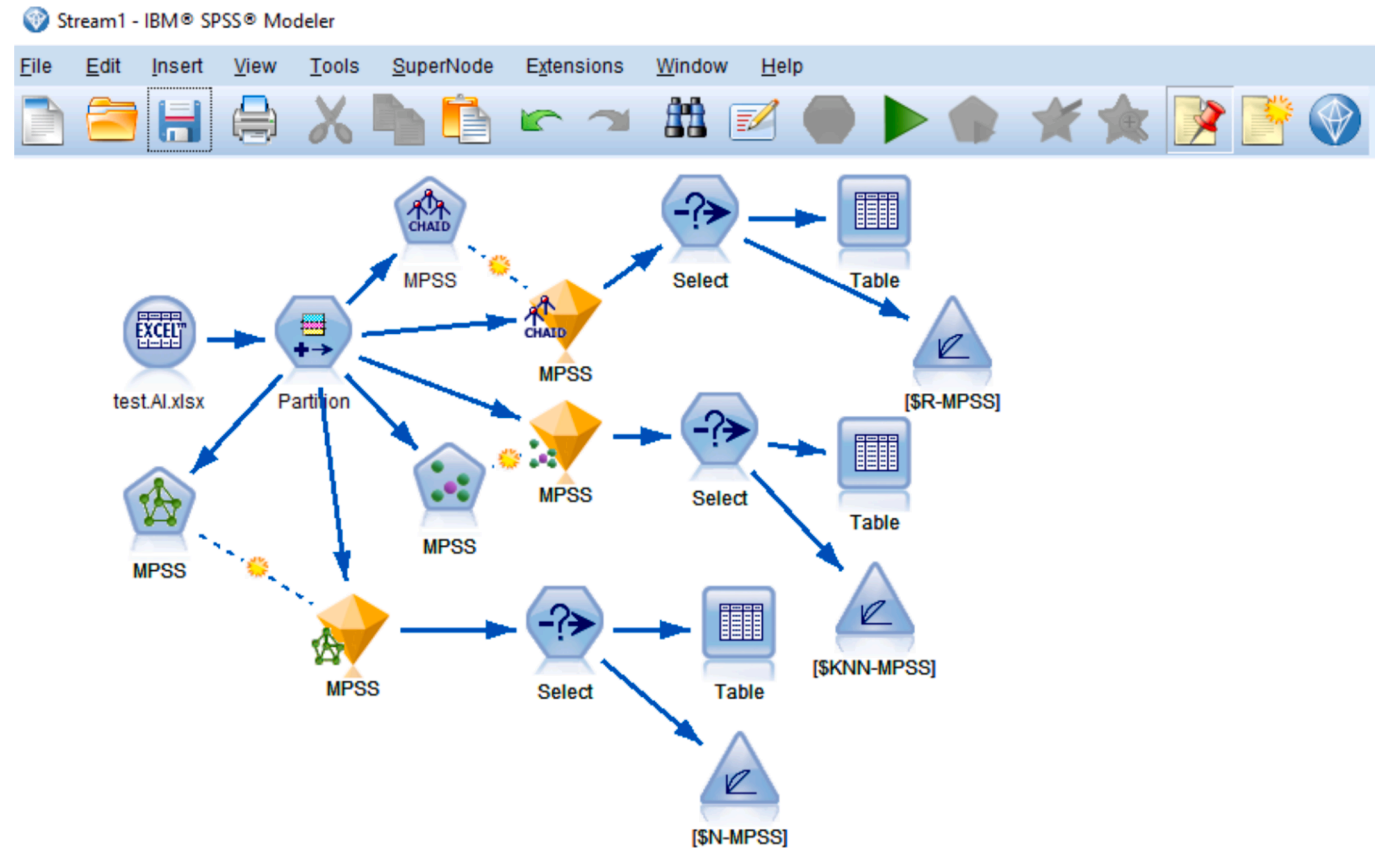


Fig. 18. The general schematic of the SPSS-modeler software

Appendix C

Fig. 19 illustrates the architecture of the Multi-Layer Perceptron (MLP) network used in the DEANN model. This structure represents a feedforward neural network with a single hidden layer comprising three neurons. Each input node—representing a supplier evaluation criterion—is fully connected to all neurons in the hidden layer, ensuring that the model captures complex nonlinear relationships among features.

The input layer includes key predictors such as CA6, CS2, DE5, DE3, DE2, QA3, CS4, CO1, DE1, CS3, R4, and CO2, along with a bias term to enhance learning flexibility. These inputs are processed through the hidden neurons using the TanH activation function, which enables the model to learn centered and normalized representations. The outputs from the hidden layer are then passed to the final node labeled MPSS, which represents the classification outcome—efficient or inefficient supplier.

This architecture was automatically optimized using SPSS Modeler, resulting in a compact yet powerful structure capable of achieving high classification accuracy (91.9 %) while maintaining computational efficiency. The fully connected design ensures that all relevant features contribute to the final decision, and the use of TanH and Softmax activation functions supports both convergence and interpretability.

DEANN network structure

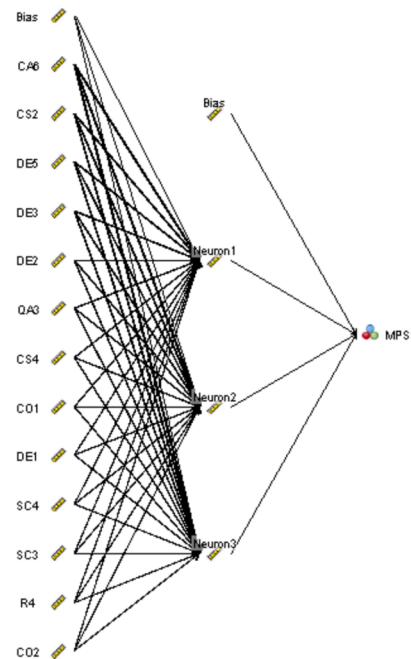


Fig. 19. The MLP structure of the DEANN algorithm

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